Castles in the Air

Extended chapter commentary

CHAPTER 8
Stairway to paradise

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CHAPTER 8 COMMENTARY

Stairway to paradise

In this chapter and the next two, we and our heroines are exploring the fundamental question:

What are the limits to what can be known in mathematics?

Are there limits? The results of Gödel and Turing would seem to imply that if our minds are basically like tube machines, then there must be mathematical truths that we can never access. But *are* our minds basically like tube machines? And if they are, is there any hope of getting some idea of where the boundaries of accessible mathematical knowledge might lie?

For the purpose of this chapter, we'll be confining our attention to what's knowable within the realm of *Arithmetic*. In the previous chapter, Emily was rooting for the idea that Peano Arithmetic provides the answer to our question: that is, she thought that the 'mathematically knowable' glyph statements were exactly those that could be proved in PA. However, this idea was rebutted at the start of the present chapter by Anna, who realized that our understanding of the principles of PA actually leads us to truths beyond the formal reach of PA itself. The girls then found themselves on a quest to discover just *how* much we can know in this way, leading them to the dizzying heights they reached in the latter half of the story.

As usual, we'll start by unpacking some of the mathematical ideas we've introduced. The two main new ideas here are the concept of *reflection* (the idea that we can recognize the truthfulness of certain theories by looking at them 'from the outside') and that of *ordinals* (the strange way of continuing numbers into the infinite which we used to label the steps of our stairway). After that, we'll look at the ideas from a more philosophical angle, and asking what light, if any, they have to shed on that deepest of mysteries: the nature of the human mind. We shall examine in detail some of the fascinating arguments that have raged back and forth on the question of whether our mathematical abilities could, in principle, be emulated by a machine in the Turing sense. Beyond a certain point, however, the discussion is in danger of losing itself in mere speculation, and at this point we shall return to the mathematics to see what further light a deeper study of logic might shed on these matters. We shall make some progress with this towards the end of this

chapter, though it will take until Chapter 9 for all the relevant ideas to come fully into view.

The role of 'understanding'

Let's start with Anna's basic insight that kicked off the chapter. If we accept all the axioms, minters and proof rules of Peano Arithmetic as being indisputably valid, then we're also naturally led to accept as indisputable the statement that *everything provable in PA is true*. And if we accept that, then it's quite easy to deduce that *PA is consistent* — a statement expressible in our original glyph language, but one which, by Gödel's second incompleteness theorem, actually lies beyond the reach of PA itself.

This may seem strange at first. We appear to have begun by committing ourselves to nothing more than our belief in PA, and then magically used this to conclude something *beyond* PA. So it's worth unpicking this a bit more.

Why do we trust PA? One way to think about this is to reflect on why you *wouldn't* trust some other random formal system I might have written down: perhaps one that extended PA with various bizarre symbols and gobbledegook axioms with no intelligible meaning. Without some understanding of what these new axioms meant, it might be hard, if not impossible, to tell whether the (intelligible) Whole number statements you could prove with them would be true ones or not. But in the case of PA, we have more than the pure symbolism to go on: we all have an *understanding* of the concept of Whole numbers — an understanding we've had since childhood — and we can readily *see* that the axioms of PA all say true things with respect to this understanding. (I'm supposing, at least for the time being, that we're not adopting a purely formalistic attitude, and in particular that our understanding leads us to accept the surveyability principle for $\mathbb{N}$ that we discussed in the last chapter.)

So a natural view to adopt is that our intuitive understanding of Whole numbers *precedes* our attempts to formalize this understanding in a system like PA, and it provides the justification for such systems. And whilst, as we've seen, PA makes a good stab at capturing the main outlines of this understanding, it turns out that it doesn't exhaust the whole of what we can 'see' to be true of Whole numbers: there are other truths which aren't captured by PA, but which are supported by just the same basic understanding as PA is, and hence carry the same level of mathematical certainty. A clear example of this is given by the principle of induction, which we wrote schematically in the last chapter as:

$$P[0] \;\&\; (\forall x\ P[x] \Rightarrow P[x+1]) \;\Rightarrow\; (\forall z\ P[z])$$

Our understanding of Whole numbers tells us that that this will be true for *any* meaningful property P that we're able to talk about. However, in a system such as PA, we can't capture the full strength of this idea — by the nature of the case, the best we can do is to say that this induction scheme holds for all properties P *expressible in the language of* PA. If we happen to extend our language so that it can express a wider range of properties P, then we'll get new instances of our induction scheme — instances that go beyond PA in what they say, but whose truth was implied all along by our original understanding of induction.

So the question naturally arises: What is the *complete* set of truths that our understanding of Whole numbers justifies? Can this itself be captured within some formal system? Or does our faculty of understanding — our ability to grasp the 'meaning' of the formal systems we use — provide an inexhaustibly fertile source of new insights that enables us to transcend any given system? This is one of the main questions we wish to investigate in this chapter.

So much for how, in principle, our understanding of PA might let us go beyond PA. But to capture this formally, we'll clearly have to add some new machinery to PA. And it turns out that a remarkable amount can be achieved by adding something that might at first seem quite innocuous: a way to talk about the *truth* of Arithmetical statements.

The truth about truth

We've seen already that our language of Arithmetic can, in some senses at least, 'talk about itself'. That is, we've seen how formulae of Arithmetic can themselves be *coded* by Whole numbers, and properties such as 'the number x codes a grammatically correct formula of Arithmetic', or even 'x codes an Arithmetic statement provable in PA', can themselves be expressed by (quite complicated) glyph formulae. So there's a natural question to ask: Can we, in a similar vein, concoct an Arithmetic formula that expresses the property: 'x codes a *true* statement of Arithmetic'?

The answer is no. Indeed, it was shown by Alfred Tarski around 1933 that *no* consistent formal system (of the sort we're considering here[1]) can express its own 'truth predicate'. This is in itself a central idea of mathematical logic, so it's worth pausing to see how Tarski proved it. For definiteness, we'll focus here on the language of Arithmetic that we're familiar with.

Suppose, then, that there *were* some Arithmetic formula T, with free variable x, that captured the property 'x codes a true Arithmetic statement'. This would mean that for any specific number n, the statement T[x→n] would be true if, and only if, n was the code for a true Arithmetic statement. But if this were so, we could use Gödel's method for constructing self-referential statements (or 'self-biting serpents') to cook up a magic number n such that

the code for the formula ¬ T[x→n] is actually n itself![2] In this case, the formula ¬ T[x→n] would effectively assert:

This statement is not true.

It's not hard to see that this yields a contradiction: if T[x→n] is true, then n codes a true statement, so ¬ T[x→n] is true and T[x→n] is false; but if T[x→n] is false, then n does not code a true statement, so ¬ T[x→n] is false and T[x→n] is true. And since we reached this contradiction from the assumption that a truth predicate T existed, we conclude that there can be no such truth predicate after all.

Incidentally, the 'self-denying' statement we've just mentioned can be seen as a variant of a very old paradox going back to Epimenides, a man from Crete who wrote that Cretans were always liars.[3] There's an interesting point worth making here: ordinary human languages, it seems, *can* quite freely refer to their own concept of 'truth', in defiance of Tarski's limitation. But the price they pay is that they're subject to paradoxes like the Epimenides one. This is because ordinary languages are not idealized formal languages in which every sentence has a precisely defined meaning — so we shouldn't be too surprised if they contain loopholes that allow paradoxes to slip in. In Tarski's precise formal setting, the 'paradox' is turned into a proof by contradiction, showing that the existence of a 'truth formula' T is an impossibility.

Adding a truth predicate

So our language of Arithmetic doesn't provide a means of talking about the concept of 'truth' for Arithmetical statements. But what if we were simply to add such a means? Since there's no *formula* T of Arithmetic that expresses the property of coding a true statement, let's simply add a new symbol **T** that does this job — so that if *t* is any term of our original language, **T**(*t*) will be a formula whose intended meaning is that the value of *t* codes a true statement of Arithmetic.

We'll also need a few new axioms to make this intention explicit. These axioms really just amount to a definition of what 'true' means, much along the lines of Tarski's own 'definition of truth' which we met in Chapter 3. Basically, we will have one axiom for each way of forming statements in Arithmetic. For example, the relevant axiom for '&' statements was mentioned in the story:

⊙ ' **T** (x & y) if and only if **T** (x) & **T** (y) '

where we understand $\underline{\&}$ as the operation on numerical codes that mirrors the & operation on formulae themselves. Of course, since $\underline{\&}$ isn't literally a symbol in our glyph language, this should be read as a shorthand for some much more long-winded glyph statement whose details I'll spare you. A similar approach can be used for statements constructed via $\neg$, $\vee$, $\Rightarrow$, $\forall$ and $\exists$, and also for basic statements constructed using $=$, $<$ and $\leq$.[4] We'll refer to the system obtained by extending PA in this way as PA_1.

What is achieved by adding this **T** operator to our system? You might at first think it wouldn't add much at all: after all, saying "The statement 'It is raining' is true" says neither more nor less than "It is raining", so it might appear that adding this concept of 'truth' doesn't really let us say anything new. It turns out, however, that PA_1 packs considerably more power than PA does — not so much because we can talk about the truth of *individual* statements, but because we can now talk about the truth of a large class of statements simultaneously.

For example, whereas in PA we needed an infinite number of axioms (which is what the minters gave us), in PA_1 it's possible to prove a *single* theorem which says, in effect,

⊙ 'All the axioms of PA are true'

And since each of our Rules of Thought take us from true premises to true conclusions, it's then not to hard to prove:

⊙ 'All theorems provable in PA are true'

As we saw in the story, we can combine this with '0 = 1 isn't true' to deduce '0 = 1 isn't provable'; and because an inconsistent system can prove anything, we conclude

⊙ 'PA is consistent'

So we can think of PA_1 as offering us a 'global' view of everything that's going on in PA simultaneously, whereas any particular proof within PA itself gives only a 'local' view of one small part of it.

Since 'PA is consistent' can be stated in the language of PA, but not proved there, we thus see that PA_1 surpasses PA in its power for proving Arithmetical statements. In this way, we can refute Emily's conjecture that every Arithmetic statement provable with mathematical certainty is provable in PA.

The hydra foiled

Are there any other interesting questions that can be settled in PA_1 but not in PA — aside from navel-gazing metamathematical questions about consistency and the like? To discover the answer, let's go back to Hercules and the hydra, whom we left locked in mortal combat in the previous chapter.

Recall that even in the basic version of their contest, Hercules seemed to be fighting a losing battle against an explosion of hydra heads which he has no hope of keeping up with. We then modified the rules so that things were even more heavily weighted in the hydra's favour. Our purpose in doing so was to formulate an seemingly modest conjecture which we might hope to be able to prove: There's some particular hydra *h* such that, for any number of rounds *r* named by Hercules, the hydra can survive for at least *r* rounds (where the hydra gets to choose at each stage which head to sacrifice). This can be framed as a statement of Arithmetic along the following lines:

$\exists h \ \forall r \ \exists b$ 'b is a battle sequence of r rounds starting with the hydra h'

What we noted in the last chapter was that (perhaps surprisingly) this statement can be neither proved nor disproved in PA.

Now what about PA_1? It turns out that with the additional power provided by the **T** predicate and truth axioms, it's now possible to settle the matter definitively. However, the answer may not be what you expect: the Arithmetic statement above can actually be proved in PA_1 to be *false*! In other words, if Hercules names a sufficiently colossal number r (which may depend on h), the hydra h will find it impossible to hold out for r rounds. Or to put it another way: in our original version of the contest, in which Hercules chooses that heads to chop, Hercules *can't help winning* provided he persists for long enough, regardless of how he makes his choices.

This may seem hard to believe, so let me see if I can make it at least vaguely plausible. (You might want to glance back at Chapter 6 to refresh your memory of how the battle works.) First of all, notice that there is at one situation, at least, in which Hercules makes a clear gain: if the head he cuts is at 'height' 1 (that is, if its neck grows directly from the root), then no new heads are grown, and the total number of heads decreases by one. So if we can eventually reduce any hydra to one in which *all* the heads have height 1, then we'll be home and dry, because Hercules can then simply pick them off one by one.

Now suppose we have a hydra in which all heads are at height 1 or 2 — that is, there are at most two segments between any head and the root. What happens if Hercules chops off a head at height 2? An enormous number of new heads might appear, but *only at height 1* — in other words, although the total number of heads might grow monstrously, the number of *height 2* heads

will go down. So Hercules could proceed by repeatedly chopping off all the height 2 heads until there are none left. Granted, this may result in a fantastical number of height 1 heads, but Hercules isn't too worried about that, as he knows that he can just pick them all off at the end.

I won't go into details just here of what happens for hydras of height more than 2, but the basic intuition is the same: if we take the view that one head higher up is 'worse' than a zillion heads lower down, then a hydra's overall badness will decrease with each round, even though the number of heads might grow terrifyingly large in the process.

Later in the chapter, we'll see what it takes to make this idea more precise. But for now, I shall simply ask you to take it on trust that this is a theorem that can be proved in PA_1, but not in PA.

The idea of reflection

To summarize, then, we've passed from our original system PA to a stronger system PA_1 by noticing that there are some legitimate kinds of reasoning about PA which we can perform by looking at PA 'from the outside' — for instance, by allowing ourselves to talk about the *truth* of PA statements — but which couldn't be performed within PA itself. The new system PA_1 was designed to capture these additional kinds of reasoning. This general procedure — of extending a system so that it can do something that could previously only be done 'from the outside' — is known as *reflection*. The idea of reflection can take many different forms, but the one that suits our purposes here consists of adding a 'truth predicate' **T** with suitable axioms.[5]

There's one very important to note, however. The truth predicate **T** that we added allows us to talk about truth for statements in the language of PA — the old language. But it doesn't let us talk about the truth of statements that themselves involve the new **T** symbol. In other words, PA_1 can talk about truth for PA, but it still can't talk about *its own* concept of truth. Indeed, Tarski's argument shows that (for the kinds of systems we're dealing with) any attempt to design a language that *could* express its own concept of truth in this way would suffer from inconsistency, since such a language would allow us build a paradoxical statement that in effect said 'This statement is not true'.

So there's still a gap in what PA_1 is able to talk about. And that leads very naturally to our next idea.

Reflecting on reflection

Having reached this point, it doesn't take too much to hit on the idea of applying this process of reflection all over again, starting from PA_1 and leading to a stronger system PA_2. This will involve adding a new symbol $\mathbf{T}_1$,

where the formula $\mathbf{T}_1(t)$ has the intended meaning 'the value of t codes a true statement in the language of PA_1'. (This of course presupposes a way of coding formulae of PA_1 as Whole numbers, but that's not hard to design.)

The new axioms for $\mathbf{T}_1$ will mostly look similar to the axioms for **T**, except that we'll now need a new axiom to cater for statements that are themselves constructed using **T**. Informally:

$$\odot \quad \text{`} \mathbf{T}_1 (\underline{\mathbf{T}}(x)) \quad \text{if and only if} \quad \mathbf{T}_1 (x) \text{'}$$

where $\underline{\mathbf{T}}$ is the operation on numerical codes that mirrors the **T** operation on formulae. Again, it might seem surprising that such an innocuous extension could lead to anything new; nonetheless, it turns out that we can go through exactly the same process as before and use PA_2 to prove 'PA_1 is consistent' — a purely Arithmetical statement (in our original glyph language!) which is beyond the reach of PA_1.

And of course, we can keep going. Allowing PA_0 as an alternative name for PA, this process of repeated reflection leads to an infinite sequence of ever stronger systems PA_0, PA_1, PA_2, PA_3, …, each able to prove the consistency of the one before. What's more, the reasoning principles captured in each of these systems have just the same claim to 'mathematical certainty' as our original system PA did, insofar as they're based on the same understanding of Whole numbers.

It's clear that this idea is very general: *whenever* we have some formal system S for mathematics whose axioms and rules we trust, we can apply reflection to obtain a system in which we can have the same kind of trust, but which goes beyond S in what it can prove. Here is how Gödel put it in a remarkable lecture given in 1951:[6]

> [The second incompleteness theorem] *makes it impossible that someone should set up a certain well-defined system of axioms and rules and consistently make the following assertion about it: All of these axioms and rules I perceive (with mathematical certitude) to be correct, and moreover I believe that they contain all of mathematics.* If someone makes such a statement he contradicts himself. For if he perceives the axioms under consideration to be correct, he also perceives (with the same certainty) that they are consistent. Hence, he has a mathematical insight not derivable from his axioms.

But that's not all. For no sooner have you grasped what I've just explained — that there's a perfectly general reflection process that gets us from the validity of each PA_n to that of PA_{n+1} — than you'll be able to see, by looking at this entire process 'from the outside', that *all* the systems obtained in this way will be valid. So we might as well collect together the axioms from all these

systems into a single big system, which will of course be more powerful than any of them individually. Even this super-system will still have a set of axioms defined in a perfectly computable way, because of the systematic, regular fashion in which the sequence of systems PA_0, PA_1, PA_2, … is defined.

We can think of what's going on here in terms of a process of repeatedly 'jumping out of the system' (an idea that plays a major role in Hofstadter's *Gödel, Escher, Bach*). That is, by continually stepping outside and 'reflecting on' the reasoning processes we are using, we are able to recognize the validity of more and more powerful systems. The first few steps of this process led us to the systems PA_1, PA_2, PA_3. But then, once we'd seen that *this* process can be continued indefinitely, we can take a bolder leap and look at the result of *that* whole process from the outside.

We can go much further than this. As we saw in the story, the general idea is that we can keep on building ever more ambitious sequences of stronger and stronger systems, but that any time we spot that what we're doing conforms to some kind of pattern or rule that could be continued indefinitely, we can 'break out' of what we're doing and jump to a higher vantage point that subsumes everything we could generate in that way. From here, of course, we can start all over again, reflecting our way onwards and upwards to greater and greater heights. This idea of trying to make an incomplete system 'more complete' by repeated reflection was first suggested by Turing in a paper from 1939.[7]

How far can this process of system-jumping be carried? Do we, in effect, have a *limitless* capacity for breaking out of any system or rule that we might appear to be following — a capacity manifesting some special creative power of the human mind? Or is there some limit to how far we can go? Or does some other factor come into play if we travel far enough? To learn the answer, we shall need to continue our journey for some distance yet; but in order to support what we are doing at these higher levels, some new mathematical ideas are needed.

We've seen that the first few stages in our upward journey can be labelled by ordinary Whole numbers (0, 1, 2, 3, …). But what comes after that? How can we even talk about the stages that come after all of these? In order to do this, we shall need to extend our ordinary idea of 'numbers' to something much more general. And this brings us to the second major mathematical concept of the chapter: the concept of *ordinals*.

On beyond infinity

We all begin by learning about the Whole numbers 0, 1, 2, 3, …. But even quite young children will sometimes ask: What comes after *all of that*? And if the parent or teacher mumbles something about 'infinity', they might ask:

What comes after infinity? Loosely speaking, you can think of ordinals as forming a 'super-number system' that supplies answers to questions like these.

I won't actually give a precise definition of an ordinal here, but will concentrate on the informal intuition. (Strictly speaking, in this chapter I'll be restricting attention to what are called the *computable ordinals*.) It's helpful to start by revisiting the ordinary sequence of Whole numbers and recalling the process by which these are generated. All we need are two rules:

1. 0 is a Whole number.
2. Given any Whole number n, there's a 'next Whole number' after n (called the *successor* of n), which we can write as $n+1$.

In a similar spirit, we can give an idea of what an ordinal is by giving the rules for generating them. There are three rules. I shall follow a common convention of using Greek letters such as α (alpha) to refer to ordinals.

1. 0 is an ordinal.
2. For any ordinal α, there is a 'next ordinal after α' (called the *successor* of α), which we can write as $\alpha+1$.
3. Whenever we can construct an increasing sequence of ordinals α_0, α_1, α_2, … (labelled by the ordinary Whole numbers 0, 1, 2, …) in some 'computable' way, there is a 'next ordinal beyond' all those in the sequence. We shall refer to this new ordinal as the *limit* of the sequence α_0, α_1, α_2, ….

The first two of these rules just mirror those for Whole numbers, but it's the third rule that's the real killer: the one that lets us 'break the infinity barrier' many times over. The way I've phrased the last rule here is pretty sloppy, because I haven't said what I mean by constructing a sequence of ordinals in a 'computable' way. To make this more precise would get a bit technical, so instead, I shall rely on examples to get the broad idea across.

Let's see what sorts of ordinals we can generate by means of our three rules, as a way of recapping on part of Anna and Emily's upward journey. Using Rules 1 and 2, we can clearly generate all the Whole numbers 0, 1, 2, …, which we may refer to these as *finite* ordinals. But these themselves form an increasing sequence of ordinals, and pretty obviously a 'computable' one at that; so we can use Rule 3 to pass to the limit of this sequence, for which we'll introduce the symbol ω (omega). We can loosely think of ω as the 'number' that comes after all of 0, 1, 2, 3, ….

Having reached ω, we can use the second rule again to carry on further: $\omega+1$, $\omega+2$ and so on. But once again, this gives us an increasing sequence of an intuitively 'computable' character, so we can invoke the third rule to pass to its limit: an ordinal we can write as $\omega+\omega$, or alternatively as $\omega\times2$. (You may

gather from this that there are certain 'rules of arithmetic' that apply to ordinals, although I won't go into these here.) Proceeding in this way, we then can go on to construct $\omega\times3$, $\omega\times4$ and so on. But since *that* is an increasing sequence constructed in a 'computable' way, we can pass to its limit, which we may write as $\omega\times\omega$, or as ω^2. And now we can start all over again, and continue up to $\omega^2+\omega^2$, which we can also write as $\omega^2\times2$, and then carry on in this way, we can construct the sequence $\omega^2\times3$, $\omega^2\times4$, …. But that is another computable increasing sequence, so we can pass to its limit $\omega^2\times\omega$, or ω^3.

At this point, however, an even bigger pattern becomes clear: the sequence ω, ω^2, ω^3,… can itself be continued indefinitely, so by taking *its* limit we can get the ordinal known as ω^ω. In fact, to cut a very long story short, we can continue this process a great deal further to obtain a sequence

$$\omega, \qquad \omega^\omega, \qquad \omega^{\omega^\omega}, \qquad \omega^{\omega^{\omega^\omega}}, \qquad \ldots$$

At this point, a change of notation is obviously called for, and the limit of *this* sequence is usually written as ε_0 — a convenient place to stop and rest for the time being. We shall return to the higher levels of the girls' ascent (including the bit about the tortoise) towards the end of the chapter.

Ordinals turn out to be a fundamental mathematical concept that crops up not just in logic, but in many other areas of mathematics where such a concept of 'super-number' is called for. However, our particular interest here is in using ordinals to extend the idea of reflection beyond the finite levels. Indeed, our three rules for generating ordinals have a natural interpretation in terms of our repeated reflection process. The ordinal 0 serves simply as a label for our starting point — the system PA_0, which we assume at the outset is truthful. Next, the 'successor rule', saying that we can always step up from an ordinal to one above it, corresponds to the fact that once we've defined a system PA_α and established it as truthful, we can always move one step up and see that the bigger system

$$PA_\alpha \ + \ (\text{truth predicate } \mathbf{T}_\alpha \text{ for } PA_\alpha, \text{ and suitable axioms})$$

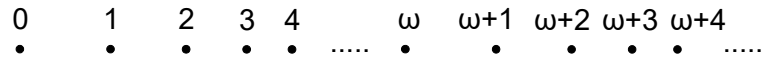
is also truthful.[8] (We allow $\mathbf{T}_0$ as an alternative name for $\mathbf{T}$.) Finally, the 'limit rule', saying that we can always pass to an ordinal beyond any given sequence, corresponds to the idea that if we have a sequence of systems PA_{α_0}, PA_{α_1}, PA_{α_2}, … all established as truthful, where α_0, α_1, α_2, … is some computable increasing sequence of ordinals, then we can combine them all into a big system PA_α, where α is the limit of α_0, α_1, α_2, …. In this way, we can keep on producing ever stronger axiom systems for just as long as we can keep generating computable ordinals using our three rules.[9]

Translating ordinal-speak into Whole number language

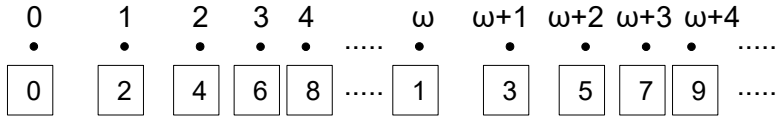
You might feel suspicious of all this talk about 'numbers beyond infinity', in just the same way that people are often suspicious of Complex numbers. Do these infinite numbers 'exist' in any reasonable sense? Or are we just weaving some curious mathematical fiction here? This is a particularly serious question in the present context: we're supposed to be investigating what can be known with *mathematical certitude*, so we don't want the waters to be muddied by dubious references to possibly fictitious entities. Even more specifically, we're in the process of investigating how far we can travel with just our basic understanding of Whole numbers, so we don't want to be forced to take on board some further concept of 'super-number' unless this concept can itself be 'explained' in terms of Whole numbers.

Fortunately, we can address this concern in a satisfactory way, because it turns out — perhaps surprisingly — that all computable ordinals can be 'represented' using nothing but ordinary Whole numbers. This might sound paradoxical at first: surely the whole point of ordinals like ω^ω is that they're much 'bigger' than Whole numbers? Yes indeed — but that doesn't stop us from creating a sort of 'picture' of an infinite ordinal using just Whole numbers. Since it's a crucial point that everything we're doing here can ultimately be grounded in our conception of Whole numbers, I'd like to give you just a glimpse of how this works — with a warning that there are some technical issues lurking here that I'll be glossing over.

To see how infinite ordinals can be represented using Whole numbers, here first is a rough picture of all the ordinals less than $\omega \times 2$ (arranged horizontally rather than vertically for convenience):



Now let's see if we can 'label' all these ordinals using Whole numbers in some way, so that every ordinal gets a different label. Obviously, if we just labelled 0 with 0, 1 with 1 and so on, then we'd have no labels left for ω, ω+1, etc. But what if we used the following alternative labelling?



Here we're using even numbers (in boxes) as labels, or *codes*, for the finite ordinals, and odd numbers as codes for the ordinals from ω onwards. Of course, this plays havoc with the usual *ordering* of the Whole numbers: for

example, 3 is less than 1000, but the ordinal coded by 3 (that is, ω+1) is much bigger than the ordinal coded by 1000 (that is, 500). But that needn't be a problem, because we can simply declare that a different kind of ordering is relevant when we're thinking of numbers as codes for ordinals. Let's define a funny alternative ordering '$\angle$' on the Whole numbers as follows:

x $\angle$ y if *either* x is even and y is odd
or x and y are both even and x<y in the usual sense
or x and y are both odd and x<y in the usual sense

The point is that this definition doesn't refer to ordinals at all, only to Whole numbers — but even so, the Whole numbers, when given the ordering '$\angle$', can be seen as a 'picture' for all the ordinals less than ω×2. In this way, a lot of things we might want to say about these ordinals can be replaced by statements about Whole numbers. For instance, if we want to define the system $PA_{\omega+23}$, we shall need — among other things — a way of coding formulae in the language of $PA_{\omega+22}$ as Whole numbers, and this language will itself involve quite a lot of truth predicates: $\mathbf{T}_0$, $\mathbf{T}_1$, $\mathbf{T}_2$, … and $\mathbf{T}_\omega$, $\mathbf{T}_{\omega+1}$, …, $\mathbf{T}_{\omega+21}$. But now that we have a way of representing all the ordinals up to ω+21 by numerical codes, it's not too hard to come up with a coding system for formulae of $PA_{\omega+22}$. In this way, many statements that appear to refer to infinite ordinals can be brought back into the fold of Whole number statements. In particular, it turns out that even something like '$PA_{\omega+23}$ is consistent' can be formulated as a statement in our *original* language of Arithmetic.

What if we want to go higher than ω×2? We can certainly do so, but we'll need to switch to a slightly more complicated code for ordinals. For instance, here's one way to code all the ordinals less than ω^2. We first notice that every such ordinal can be written in the form ω×m + n, where m and n are ordinary Whole numbers. But we've also seen how a *pair* of Whole numbers m,n can be coded by a single Whole number p, so we could simply use this number p as our code for the ordinal ω×m + n. (For instance, using the 'alternating digits' coding of pairs we saw in Chapter 4, the ordinal ω×23 + 71 would be coded by the number 2731.) Using this trick, we're able to define the system PA_α for any ordinal ω^2 below .

Of course, once we reach ω^2, even this code will 'run out', and we'll have to invent a yet more elaborate one to allow us to go even further. In fact, it turns out that *any* computable ordinal can be depicted using *some* coding system of this kind — and what's more, the relevant funny ordering '$\angle$' associated with the coding can be defined in a concrete, computable way that refers only to Whole numbers.

This continual switching from one coding scheme to another might seem a bit annoying. Why don't we simply fix once for all on some coding that's

powerful enough to represent *all* the computable ordinals, so that we don't have to keep chopping and changing as we go up to larger and larger ordinals?

Unfortunately, it turns out that the chopping and changing is unavoidable. Although every ordinal can be represented using *some* suitable coding, there's no *single* coding system (of the sort we're talking about here) that's capable of representing *all* the computable ordinals. Any coding system you think of will only take you so far, and then you'll have to move to a different one in order to carry on: indeed, the coding system itself will point the way to a computable ordinal that lies beyond its own reach.

This is all perfectly consistent with our earlier idea of repeatedly 'jumping out of the system' in order to reach higher and higher levels. We can loosely think of a particular coding convention as a sort of 'system' or 'framework' for talking about certain ordinals. But whenever we set up such a framework, the mere fact of doing so means that we can construct an increasing sequence of ordinals that eventually surpasses any given ordinal within the framework, and so by Rule 3 we can jump to an even bigger ordinal that lies beyond the framework. This might lead us to design a broader framework encompassing even more ordinals, but this too will have its limits and can be transcended in its turn.

A closely related point is one that Anna touched on with her suggestion that climbing up the ordinals somehow requires *creativity*. Indeed, no purely mechanical process for producing a sequence of ordinals will be able to reach 'all the way' up the computable ordinals (in the sense that every computable ordinal will eventually be surpassed by one in the sequence) — precisely because the *limit* of any such sequence would itself be a computable ordinal beyond the sequence's reach. So *if* there is to be some process generating a sequence of ordinals that goes 'all the way' in this sense, it will certainly have to be doing something non-computable.

Minds versus machines

At this point, it's natural to slide into a more philosophical gear, and consider what all of this might or might not have to say about the vexed question that Anna and Emily are debating: Could the functions of the human mind in principle be replicated by some purely mechanical or algorithmic process?

We're all familiar with science fiction books and films that feature robots and computers possessing all the faculties of human intelligence, and often much more. But is this just fantasy, or does it represent a genuine possibility? Might the future development of technology really reach a point where everything our human minds can do could be simulated by a computer? And even supposing one could construct a computer that was indistinguishable

from a human in terms of its outwardly visible *behaviour*, could a computer really have the same kind of inner subjective states that we have — the conscious experiences of perceiving, thinking, feeling and suffering that we associate with being human? Are such questions even sufficiently precise to have straight yes-no answers?

Questions like this have formed the subject of a vast, complex and often heated debate, ever since the French philosopher Julien Offroy de La Mettrie published his controversial book *L'homme machine* in 1747. We shall be confining our attention here to just one small part of this debate: namely, the question of whether our human capacity for *mathematical* thought, and mathematical understanding, is something that could in principle be equally well possessed by a mechanical computer. Even more specifically, we'll be considering whether the mathematical ideas we've just been discussing have any bearing on this question.

To begin rather loosely, one can find oneself thinking the following kinds of thoughts. We've seen that any given 'mechanical' process can only go so far in the computable ordinals it can generate before running out of steam. So *if* the activity of our minds can in principle be modelled by some mechanical process, surely it follows that there are levels in the ordinal staircase that we'll never be able to 'reach' — and hence, perhaps, axiom systems whose truthfulness we'll never be able to perceive? On the other hand, if there's something irreducibly non-mechanical and 'creative' about our capacity for jumping out of the system — if the human mind can break out of any cage in which one tries to contain it — then it might seem quite on the cards that *any* computable level of the staircase is in principle accessible to us, and any of the corresponding axiom systems can be seen to be valid. (The phrase 'in principle' here should be understood to mean something like 'ignoring limitations on human memory, lifespan, perseverance and the like'.) Clearly, these alternatives offer us two quite different visions of what the human mind is like, and it's natural to ask which is the more believable.

Gödel's disjunction

Actually, the central point here can be put much more simply and crisply, and we have already met it in Chapter 5. We know that no tube machine is capable of sorting Arithmetic statements into two piles: the true ones and the false ones. So *if* we can be simulated by a tube machine (as far as our capacity to perceive mathematical truth is concerned), it follows that *we* can't sort glyph statements into true and false ones either — that is, there must be Arithmetic statements whose truth is in principle *unknowable* to us. What our discussion of computable ordinals now adds to this picture is the suggestion

that in this latter case, the ladder of ordinals might serve as some sort of 'scale' for measuring just *how* much it's possible for us to know.

The basic conundrum we have just outlined is often referred to as *Gödel's disjunction* or *Gödel's dichotomy*. It was formulated by him as follows in the 1951 lecture we cited earlier:[10]

> For if the human mind were equivalent to a finite machine, then … there would exist absolutely unsolvable diophantine problems … , where the epithet "absolutely" means that they would be undecidable, not just within some particular axiom system, but by *any* mathematical proof the human mind can conceive. So the following … conclusion is inevitable: *Either ... the human mind (even within the realm of pure mathematics) infinitely surpasses the powers of any finite machine, or else there exist absolutely unsolvable diophantine problems*.

Gödel goes on to say that strictly there are three possibilities, since it's conceivable that *both* of these alternatives might hold. 'Diophantine' problems were something we touched on briefly in Chapter 4 — Gödel mentions these here to emphasize that the unsolvable problems in question could be chosen to be of an extremely simple kind. In particular, Diophantine statements are examples of what we called *Predictive* statements in Chapter 5 — as also are the consistency statements featuring in the second incompleteness theorem. In the present chapter, however, we need not be too concerned with such refinements, since we're working under the background assumption that *all* statements of Arithmetic are meaningful. We shall therefore be content to replace Gödel's second alternative by the weaker claim: 'there exist absolutely unknowable statements of Arithmetic'.

Was Gödel right? Is it indeed inescapable that at least one of the above alternatives holds? Is Gödel's disjunction merely a claim about some theoretical idealization of our mathematical ability, or does it have anything of interest to say about the actual capacities of real flesh-and-blood mathematicians? If it does, then can one go further and argue for or against one of the alternatives? In particular, do Gödel's theorems lend support, as some have claimed, to the first alternative of his disjunction — to the idea that there's something in our minds that surpasses the power of any machine?

These are some of the questions we shall be exploring over the next few pages. The whole area is still highly controversial, and we shall spend some time batting around some of the points that have arisen in the course of discussions. We'll focus first on Gödel's disjunction itself, then move on to some the arguments — specifically those of the mathematician and physicist Roger Penrose — that have been put forward in support of the 'non-mechanist' alternative. At a certain point, however, we shall find that we need to resume our exploration of mathematical logic to gain a fuller picture of

what is going on, and at this point we shall return to the latter part of the girls' ascent.

Gödel's disjunction scrutinized

Let's think a bit more about Gödel's disjunction in the carefully worded form which we quoted above. On one level, this certainly *seems* to follow from Turing's results: no machine can answer all mathematical questions, so either there are mathematical questions *we* can't answer, or we must have some ability that can't be replicated by any machine. Surely, the reasoning here is at least broadly correct.

On the other hand, one might point out that Gödel's statement refers to various concepts that aren't defined with complete precision: for instance, that of 'the human mind' and that of an 'absolutely unsolvable problem'. Can these concepts be made clear enough for the disjunction to say something substantial and interesting?

Let's start with 'the human mind'. *Whose* human mind are we talking about? Or are we really talking here about the power of 'human reason' in some purified or idealized sense, abstracting away from limitations of time and energy — not to mention that tiresome human propensity to make mistakes with disturbing frequency? Can this rather vague concept of an 'idealized mind' be brought into sharper focus? And if it can, what would it mean for such a mind to 'surpass the powers' of a machine? What standard of comparison are we using?

Then there's the concept of an 'absolutely unsolvable problem'. Gödel says that this means a problem not solvable 'by any mathematical proof the mind can conceive' — but is it sufficiently clear what we mean by this? After all, the question of what counts as a 'proof' is one of the main things we are investigating in this book.

Of course, it needn't be that all these concepts have to be made *completely* precise for Gödel's argument to have some genuine force. But it's nonetheless interesting to see just *how* clear we can make them, especially if we're wanting to consider which of Gödel's alternatives seems the more plausible. And it seems to be agreed that *if* can formulate all the concepts with sufficient clarity, then Gödel's disjunction will indeed follow inescapably.[11]

There's a certain amount of fun to be had in trying to tighten up these concepts so as to nail down the disjunction to a more precise claim, and trying to anticipate the various quibbles that might arise. So let's spend a bit of time making a stab at this, to see how close we can come to a watertight (but still interesting) statement. I shall draw here on various points made in the research literature,[12] adding some minor twists of my own. However, this kind of

discussion is admittedly not to everyone's taste, so don't feel obliged to follow all the arguments in detail.

As a first move, I propose that we recast Gödel's disjunction just slightly into the following 'implicational' form:

> *If* the human ability to identify certain Arithmetic statements as 'true with mathematical certainty' can be simulated by a machine, *then* there are statements of Arithmetic that humans cannot resolve with that same mathematical certainty.

I'll take the liberty of referring to this as *Gödel's implication*.[13] Although it may appear equivalent to the disjunction, it has the following advantage: whereas it might in general be very difficult to make precise the concept of a 'humanly resolvable' statement of Arithmetic, things get a lot easier if we're only required to do this *under the hypothesis* that humans can be adequately simulated by machines. On the other hand, if we find that we have trouble in making this 'mechanist hypothesis' sufficiently precise to count as true or false, then the implication form will not claim anything, so in this respect it will be slightly weaker than the disjunctive form.

Now let's see just how far we *can* tighten up the mechanist hypothesis.

As a start, taking our cue from the 'clerk-in-a-box' model from Chapter 4, one might devise thought-experiments involving some specific human mathematician H (female on this occasion) isolated in a room or flat, communicating with the outside world only by email, in order that it should be clear what it would mean for H's possible 'observable behaviours' to be mimicked by a machine. We may suppose that H's job is to 'publish' theorems once she is satisfied that they are true with mathematical certainty (this could be done by emailing formal glyph statements of the theorems to some specified address). Note that H doesn't necessarily have to come up with the proofs of these theorems herself: she is free to receive hints, or even detailed 'proofs', from the outside world. All she has to do is to judge whether the arguments put forward for some theorem indeed amount to an absolutely unimpeachable mathematical proof. The idea is then that if H were indeed equivalent (in emailing behaviour) to some machine M, then the set of theorems that H could potentially publish (given suitable inputs from the outside world) would be precisely defined, as the set of theorems publishable by M would be precisely defined. This suggests the following 'next approximation' to Gödel's implication:

> If H's ability to identify certain Arithmetic statements as 'true with mathematical certainty' (under the proposed experimental conditions) can be simulated by a machine M, then there are

> statements of Arithmetic which H could not resolve with that same mathematical certainty.

However, there's already a problem here. If H is simply a human with some finite lifespan, then surely it's *trivially* true that H's can be mimicked by a machine — if only because the total number of lifetime email exchanges that H could potentially have is finite (albeit ridiculously large)?

To pre-empt this trivialization, I suggest the following move. We can note that when the proponents of Artificial Intelligence (for instance) claim that human minds can be simulated by machines, they surely mean to say something more than that a particular finite set of finite behaviours could theoretically be pre-programmed into some ridiculously huge machine (probably much bigger than the known universe). If the advocates of AI weren't prepared to put *some* upper bound on the size of the required machines — one that isn't so absurdly large as to make their claim vacuous — then no one would be impressed by their claims. Typically, however, such people are (at least implicitly) claiming that human minds could be adequately mimicked by machines of a kind that we might one day be able to build (perhaps by some sophisticated bootstrapping process involving machine learning, artificial evolution and the like). This suggests that it might be more interesting to consider a variant of Gödel's implication along the following lines:

> If H's ability to identify certain Arithmetic statements as 'true with mathematical certainty' (under the proposed conditions) can be simulated by a *feasibly constructible* machine, then there are *feasibly constructible* statements of Arithmetic which H could not resolve with that same mathematical certainty.

The phrase 'feasibly constructible' is intentionally open-ended here: we could make the statement more precise simply by choosing some limit to the size of a machine one would accept as 'feasible'. The point is that given any 'feasible' machine M, the size of the Arithmetic statement expressing that 'the outputs of M are consistent' will be only just a bit bigger than M itself, so can still be considered feasible.[14] We would thus obtain a whole spectrum of versions of Gödel's implication, clarifying that the bolder the 'mechanist thesis' (i.e. the smaller the machine is hypothesized to be), the stronger the 'unknowability' conclusion. This also, incidentally, deals with the worry that the conclusion might hold *trivially*, simply because there are statements too large and complex ever to be constructed in practice.

One might still feel, though, that the conclusion is too weak to be of interest, since one wouldn't in any case expect that any 'feasibly constructible' Arithmetic statement would be resolvable within the lifetime of the single

mathematician H (let alone within the time for which she is willing to put up with this state of solitary confinement). For this reason, and perhaps for others too, we might at this point switch to the image of a whole community or hive of such mathematicians sitting in little hexagonal cells, freely communicating by email with one another as well as with the outside world. Once again, the job of the community is to 'publish' theorems once they are satisfied that they are true with mathematical certainty — in this they may benefit from hints from the outside world and from from discussion among themselves. The idea is then that if each of our mathematicians *separately* were equivalent (as regards email behaviour) to some feasibly constructible machine, we could again speak quite precisely of the set of all theorems that this community of mathematicians could potentially publish. What's more, it would then become *mathematically* meaningful to ask how the community of such machines would behave if its existence were prolonged infinitely into the future: for instance, by repeatedly introducing 'clones' of our machines into the community to replace those that have expired. Such a community could then work over several generations on mathematical problems whose mere statements were too complex to be apprehended by any single mathematician within their lifetime. Even if the problem at hand is to understand the workings of some colossal machine M that's too large to be grasped by any one individual, we can still ask whether a sufficiently large-scale community research project could ever attain such an understanding. (This seems, in any case, closer in spirit to the question of whether there are absolutely unsolvable problems: we're interested in what *we as a race* could potentially know, given enough time, not just in what one individual can ascertain.) So we're led to the following version of the implication:

> If each of some finite set of mathematicians can be simulated (as regards their email behaviour) by some feasibly constructible machine, then there are feasibly constructible statements of Arithmetic that could never be resolved by any infinitely prolonged community constructed (in a computable way) from clones of these finitely many machines.

Clearly, this is still not a *completely* watertight statement — there are further details here to be firmed up, and further quibbles one might raise — but perhaps it's now starting to feel as though we have something solid enough to work with, and have addressed at least some of the main worries over 'idealization'. One can continue in this vein, trying to anticipate various objections and deal with them[15] — although it's all too easy to get embroiled in endless nit-picking and counter-nit-picking, and one has to draw a line somewhere. I shall therefore not pursue this avenue any further, except to discuss one very important issue that has been emphasized by a number of

commentators: the highly lamentable but undeniable fact that humans make mistakes!

Fallible humans, fallible machines

The basic problem is this. No one disputes that humans are prone to making mistakes, and there's a strong feeling that whatever safeguards we tried to put in place, some errors would be bound to slip through. So in the thought-experiment we've described, it would seem over-optimistic to assume that our mathematical beehive never publishes any erroneous theorems: if we want to be saying something about the capabilities of *real* humans, we have to take account of the possibility of mistakes creeping in.

Let's suppose, for the sake of argument, that we're clear on what it would mean for a machine to simulate (the email behaviour of) a human mathematician. Such a machine would then simulate any mistakes the human might occasionally make, and the machine would be no more reliable than the human. So although, on this hypothesis, there might be some precisely defined set of 'theorems the community could potentially publish', it couldn't be guaranteed that all these theorems were actually *true* — or even that this set of theorems was *consistent*. But if we don't know that we're talking about consistent sets of theorems, then Gödel's theorems don't even kick in, and they tell us nothing about what's going on. Essentially this point was made by Turing in a lecture in 1947:[16]

> I would say that fair play must be given to the machine … if a machine is expected to be infallible, it cannot also be intelligent. There are several mathematical theorems which say almost exactly that. But these theorems say nothing about how much intelligence may be displayed if a machine makes no pretence at infallibility.

The point as it stands is indisputable, and we have to take it seriously. Of course, what we'd *like* to say here is that we're not talking about the theorems that our community of mathematicians actually does produce, but about the theorems they *ought to* produce if they were living up to their own ideals of rigour and correctness. But can this informal idea be nailed down to something sufficiently precise for the incompleteness theorems to get into gear? I'd like to suggest two lines of response to this challenge.

The first takes off from a point often made by Roger Penrose in this context: mistakes are *correctable*, in the sense that mathematicians can recognize them as such if they are pointed out. This suggests that we might modify our thought-experiment in something like the following way. Whenever our infinitely prolonged machine community wishes to publish a theorem, it must first pass through a review phase, during which an external

'critic' has the opportunity to challenge the purported argument. If mistakes come to light, then the community can withdraw the claimed theorem. (This is in principle not dissimilar to the way the peer review process works for published mathematical papers.)

Obviously, this move doesn't buy us anything if it requires us to assume the existence of an infinitely intelligent and infallible critic. But actually, we don't need to assume this — indeed, it may be as well to think of the critic as just a monkey at a typewriter, because the idea is to consider the totality of *all possible* review messages that the monkey critic might potentially type, and ask how the mathematical community would respond in each case. (This will indeed be a well-posed question if we are indeed talking about a community of machines.) Note that we're not asking our mathematician-machines to wade their way through some ludicrous number of reviews; rather, for each possible review, we're considering *separately* the scenario in which the community receives that particular review. Of course, the vast majority of potential 'reviews' will consist of mere gobbledegook, and will be discarded by the mathematicians after a brief glance. But if there is indeed an error in the proof, then *somewhere*, amidst this immense ocean of gibberish, there will surely be a lucid, beautifully crafted message that precisely pinpoints the problem in exactly those terms that the mathematicians are most likely to understand. This will then cause them (perhaps after some internal discussion) to withdraw their theorem and go back to the drawing board. The idea, then, is that we can now define the set of 'theorems knowable by our machine community' to be the (still precisely defined) set of those statements which they might propose as theorems *and which no possible review would persuade them to retract*.

One can flesh out the details of this idea in a various ways, depending (for instance) on exactly how one wants the review process to work; and for each of these one can formulate a corresponding variant of Gödel's implication.[17] Doubtless there are things in this setup one might still quibble over, but nonetheless, it seems to me that it goes a long way towards allaying the original concern. Whilst it's commonplace to point out that mathematicians make mistakes, it would be quite radical, I think, to maintain that a community of mathematicians could collectively make some mathematical mistake *which no possible explanation could lead them to see*. If that were the case, there would surely have to be something very weird going on indeed.

A second possible response to the 'mistakes' issue has more of the character of a plausibility argument, and takes a slightly different tack. Rather than sticking with our machine model of the human mathematicians as they are and adding a refinement to address the possibility of errors, we can ask about the prospects for 'idealizing' slightly and replace our machines by ones that *didn't* make mistakes (and hence would no longer exactly correspond to real humans in their behaviour), but were in other respects 'like us'. If this were possible,

then the mathematical capabilities of these 'cleaned-up quasi-humans' would be something to which Gödel's theorems would apply.

As many people have pointed out, the problem is that it's highly unclear what behaving 'just like us, but without the mistakes' might mean. Even if we had a machine that behaved just like us, it's not clear in what way we'd have to tweak it so as to eliminate the mistakes. In other words, we simply don't know how we would come up with a suitable 'idealized' model of our mathematical abilities, abstracting away from our human fallibility.

On the other hand, it would seem rather bold to assert that *no* such idealization is or will ever be possible, under any conceivable future advances in our understanding of how brains work. Indeed, this would seem tantamount to saying that our propensity for mistakes is actually *inextricable from* our ability to perceive certain mathematical truth, and there's something about this that many find distinctly implausible. Let me unpack this a bit further.

Let's return for a moment to the alien visitors we envisaged in Chapter 1, who are trying to understand, through a detailed analysis of our brain physiology, how it is that we humans have the ability for mathematical perception that we do. Naturally, the aliens are aware that we frequently make mistakes; but they are simply looking for a satisfactory explanation, on an ordinary scientific level, for how and why the 'human proof engine' works as well as it does. Let us suppose, furthermore, that they actually succeed in *finding* such an explanation.

My suggestion is that, in our ordinary experience of understanding 'how things work', such an understanding also leads us to a clear view of the conditions under which it will *work correctly*. For instance, in a car there are several things that might go wrong; but once we understand how the car works, we are in a position to say that *if* each of the parts functions in a certain intended way, then the car will go. In other words, our understanding gives us what we would need to build an idealized mathematical model of a *correctly* functioning car (and of the physics it relies on), and even to *prove* that such an idealized car would work as intended. The thought, then, is that it would be strange if this were not also the case for the human proof engine, once it had been sufficiently analysed: if we satisfactorily understood *how* we perceive mathematical truth, wouldn't this also tell us the conditions under which we do so correctly? And wouldn't this lead us to some appropriate idealization of our mathematical capabilities to which Gödel's theorems would apply?

If this were not the case, then as I have suggested, we would seem to be led to the conclusion that our very imperfection is in some way essential to, and indeed ineradicable from, our capacity to perceive certain mathematical truth. There are indeed some who have suggested this, and the idea might even seem to have a certain attraction to it. It seems to me, however, that this response will be an unsatisfying one unless someone can (in due course) come up with

a convincing explanation of *how* our mathematical faculty depends essentially on our propensity for errors.

I've devoted some space to this issue of 'mistakes' as it has loomed large in discussions of Gödel's disjunction — and indeed of Penrose's arguments which we'll come on to later — and some commentators have seen this issue as a deep and fundamental flaw in such lines of argumentation. I completely agree that this issue of fallibility needs to be taken seriously, but I've tried to argue here, at the very least, that it's not *obvious* whether this issue amounts to a complete show-stopper, or whether it can be addressed in some way. Although neither of the above lines of response amounts to a perfectly rigorous argument, it seems to me that they take most of the sting out of the problem. So I would suggest that it's precarious to put *too* much weight on the 'fallibility' objection in these discussions.

What's more, even if the fallibility issue ultimately *is* an essential and ineradicable one, I would suggest that it doesn't offer the deepest or most satisfying insight that there is to be had on what's really going on. To be too quick to resort to the fallibility issue as a 'let-out', and leave it at that, would be to risk missing out on something far more interesting and fundamental. I hope that this claim will appear justified once our bigger picture has been uncovered.

The disjunction broadly accepted

We've been exploring how close we can come to formulating Gödel's disjunction (or implication) as a completely precise statement that follows ineluctably from the incompleteness theorems. In the above discussion, I've tried to show that one can make a pretty fair stab at addressing the main concerns and bringing Gödel's claim into sharper focus.

Actually, there's one very important question that I deliberately *haven't* addressed in this discussion: that of whether it's sufficiently clear what we mean by 'knowing with mathematical certainty'. I've refrained from doing so here because this is, in essence, the very question that we are in the course of investigating in this and the next few chapters. For the time being, suffice it to say that everything we've been saying so far ought to be true under *any* reasonable way of making the concept of 'mathematical certainty' precise. But a fuller picture of what is going on here will emerge in due course.

Aside from this ongoing question, how successful has our attempt at precision been? I certainly don't claim that we've succeeded in rendering Gödel's claim *absolutely* watertight. But there again, such 'absolute watertightness' is rarely if ever achievable by any argument in philosophy or even in natural science, and we shouldn't dismiss arguments simply on that account. So the right question to ask, perhaps, is whether Gödel's disjunctive

claim carries sufficient force to be broadly accepted as a basis for further inquiry.

In the light of our discussion, it seems clear to me that the core of the idea is perfectly secure and robust enough that it establishes *something* non-trivial. In particular, I don't think anyone today would maintain that the statements 'Human minds can be mimicked by machines' and 'All problems of Arithmetic are humanly solvable' can both be made clear enough to have a definite truth value, and that both are true! Over the last few pages, I've furthermore tried to argue that if the first claim can be made clear enough to be true, then the second can be made clear enough to be false. Likewise, I think one can reasonably argue that if the second is clear and true then the first is clear and false. Perhaps not all will agree with me on these points; but I think most people would allow that at least *some* progress is made by Gödel's disjunction, and would concur with Solomon Feferman's careful verdict:[18]

> What I have been concerned with … is to put into question Gödel's assertion that his dichotomy is an established mathematical fact. But at an informal, non-mathematical, more every-day level, there is nevertheless something to the ideas involved and something to the argument that we can and should take seriously.

For the rest of this chapter, then, I propose that we do indeed take Gödel's disjunction seriously, if only to see where it leads. But this, of course, brings us to the Big Question. *Which* side of the disjunction is the more plausible? Might they *both* be right? Or is it rather that one or both of our propositions are irremediably imprecise, despite our best efforts at tightening things up?

A sample of views

Both of the dilemmas raised by Gödel's disjunction — whether our minds can be adequately simulated by machines, and whether there are mathematical questions whose answer lies fundamentally beyond our reach — are clearly huge philosophical conundrums, and both have been the subject of deeply held convictions and sometimes heated debates. Before we look at some of the arguments, let's take a quick sample of the positions held by some of the figures who have played major roles in the development of the subject. Speaking for myself, I find it a perpetual source of fascination that these deep and original thinkers — all of whom had a profound grasp of the mathematical ideas in question, were well acquainted with the scientific knowledge of the time, and were perfectly capable of swimming against the tide of received opinion when necessary — should have arrived at such radically different conclusions.

Since we're only going here for a very broad impression of the spread of opinions, I shall crudely treat each of the dilemmas here as a straight choice between two alternatives, thereby shamelessly obliterating most of the subtleties I've been carefully drawing attention to so far. Taking this coarse-grained approach, the following grid shows the possible combinations:

	All questions are answerable	There are unanswerable questions
Minds are like machines	*ruled out*	**mechanism**
Minds surpass machines	**optimism**	**between-ism**

The combination marked *ruled out* is the one excluded by Gödel's disjunction, and as I've said above, I don't know of anyone today who would actively contend for it. But the other three positions are all consistent with it. The broad view that minds can be simulated by machines is often referred to in the literature as *mechanism* (or sometimes as *computationalism*); likewise, the view that all (mathematical or scientific) questions are in principle answerable is often known as *rationalistic optimism*, here abbreviated to *optimism*. In our story, Emily is rooting for the mechanist view, and Anna for the optimist one.

I've chosen *between-ism* as a name for the final alternative as I can't think of anything better: on this view, the power of the mind would (loosely speaking) be more than that of any machine, but less than what would be needed to answer all mathematical questions. One could, of course, add a third row and column for the possibility that the relevant question is too vague to have a definite answer, but then I'd have to dream up names for all the other positions. (Nonetheless, you might like to ponder where in such a 3×3 grid you might place yourself.)

Let's begin with Hilbert. At least before Gödel came along, Hilbert's stated convictions would have put him squarely in the *ruled out* box: he wanted to have his cake and eat it. On the one hand, as we've seen in earlier chapters, Hilbert fully expected that *all* mathematical questions would be answerable by a single machine or algorithm; and if this were true, then of course, such a machine could do everything that we could. Of course, this particular conjecture was refuted hands-down by Turing's results. But even aside from this, Hilbert held a very deep belief in the solvability (in principle) of every mathematical problem. In his historic 1900 manifesto for the future of mathematics, he put it like this:

> Is it possibly a general law inherent in the nature of the mind, that all questions which it asks must be answerable? … This conviction of the solvability of every mathematical problem is a powerful incentive to the worker. We hear within us the perpetual call: There is the problem. Seek its solution. You can find it by pure reason, for in mathematics there is no *ignorabimus*.

Years later, in 1930, Hilbert reiterated this point with a more polemical edge:

> For the mathematician there is no Ignorabimus, and, in my opinion, not for natural science either. In an effort to give an example of an unsolvable problem, the philosopher Comte once said that science would never succeed in ascertaining the chemical composition of the stars. A few years later this problem was solved … The true reason why Comte could not find an unsolvable problem is, in my opinion, that there is *no* unsolvable problem. In contrast to the foolish Ignorabimus, our credo avers: We must know, We shall know.[19]

As we shall see below, Hilbert does not stand alone in this deeply held belief that *no* truly meaningful questions are in principle beyond the reach of the human intellect — a mindset to which any suggestion of a limit to the competence of human reason appears as utterly unacceptable.

How did Hilbert's position develop after the epoch-making discoveries of Gödel and Turing? Sadly, we don't really know, because of his near-total silence on the subject once Gödel had published his results. According to his colleague Paul Bernays, Gödel's theorems made Hilbert very angry, and it seems that he never really acknowledged what they implied regarding the downfall of his own programme. In view of the passionate strength of both his earlier convictions, one can only surmise that these discoveries must have come as a deep psychological blow to him.

What were Gödel's own views? In his published works he was typically very guarded and cautious in his philosophical remarks, and somewhat reticent about his personal convictions. However, his private correspondence and conversations reveal him to have been clearly within the 'optimist' camp. On the one hand, he shared Hilbert's deep faith in the unlimited power of human reason. On the other, he was also a convinced anti-mechanist of a very radical kind indeed, holding that the mind was in fact something quite distinct from the physical brain, and rejecting the contrary idea as merely 'a prejudice of our time'. Gödel also thought, incidentally, that *either* alternative in his disjunction would have implications that were 'very decidedly opposed to materialistic philosophy': in particular, the proposition that mathematical truth exceeds knowability seemed to him to imply that there was something in

mathematics that was not merely of our own creation. We will return to this intriguing idea in Chapter 10.

Next up Turing. He is often thought of as the archetypal mechanist, and many of his mature statements indeed support this view. He is well-known for devising the 'Turing test' — a thought experiment in which (in effect) an observer attempts to distinguish between a human and a machine via their email behaviour — and he fully expected that in due course we would be able to build machines that could pass for a human under this test. In a talk given in 1951, Turing commented:

> It is customary … to offer a grain of comfort, in the form of a statement that some peculiarly human characteristic could never be imitated by a machine. I cannot offer any such comfort, for I believe that no such bounds can be set.

The traditional characterization of Turing as a mechanist seems basically fair, though he also showed at times that his mind was by no means closed to other points of view.[20] A more unequivocal representative of mechanism would be Marvin Minsky — one of the pioneers of Artificial Intelligence, and the originator of our tube machines — who famously claimed that a human brain was nothing but a 'computer made of meat'.

I'll turn next to Solomon Feferman, the co-discoverer of some of the fascinating properties of the 'ordinal staircase' that emerged towards the end of our story (and which we'll come onto soon). We may classify Feferman as a cautious occupant of the between-ist position: 'Speaking for myself, I have to say that I find it plausible that *both* disjuncts of Gödel's dichotomy are true!'. On the one hand, Feferman declared that he was 'personally convinced of the extreme implausibility of a computational model of the mind'; on the other, he was willing to countenance the idea that a question like 'What is the $10^{10^{10^{10}}}$th digit in the decimal expansion of π?' might never be answerable in any practical sense.[21]

A few other views are worth sampling. The logician George Boolos suggests that the claim that there are unanswerable questions is not only plausible but 'does not seem particularly surprising or remarkable at present' — a viewpoint that appears to be quite popular today. However, Boolos does concede that 'it may be precisely because we have come to understand the incompleteness theorems so well' that we find this idea unremarkable! The distinguished mathematician and philosopher Hilary Putnam speaks for many, I think, in maintaining that the supposed 'set of humanly provable theorems' is just too imprecisely defined for us to ask whether it's computably generated or not. (Our earlier discussion of Gödel's disjunction can be read as a stab at responding to some of this concern.) And the logician Peter Koellner, at the end of a very thorough discussion, offers an intriguing conclusion:

> *Either the statements that "the mind cannot be mechanized" and "there are absolutely undecidable statements" are indefinite ... or they are definite and the above results and considerations provide evidence that they are about as good examples of "absolutely undecidable" propositions as one might find.*[22]

To conclude our straw poll, let us now turn to the mathematical physicist Roger Penrose. It's particularly interesting here to read Penrose's account of some personal history from his days as a graduate student:

> I first heard about the details of Gödel's theorem as part of a course … given by the Cambridge logician Steen. … I had vaguely heard of Gödel's theorem prior to that time, and had been a little unsettled by my impressions of it. My viewpoint before that would probably have been rather close to what we now call 'strong AI' [a mechanist view]. However, I had been disturbed by the possibility that there might be true mathematical propositions that were in principle inaccessible to human reason. Upon learning the true form of Gödel's theorem … I was enormously gratified to hear that it asserted no such thing; for it established, instead, that the powers of human reason could not be limited to any accepted preassigned system of formalized rules. What Gödel showed was how to transcend any such system of rules, so long as those rules could themselves be trusted.

This quotation shows Penrose to have been initially both a mechanist and a staunch rational optimist. However, he goes on to say how Gödel's theorems forced him to change his position:

> My reason for presenting this bit of personal history is … to demonstrate that … the Gödel argument was already strong enough to turn at least one strong-AI supporter away from computationalism. It was not a question of looking for support for a previously held "mystical" standpoint … But the very force of Gödel's logic was sufficient to turn me from the computational standpoint with regard not only to human mentality, but also to the very workings of the physical universe.[23]

With Gödel, then, Penrose can be classified as an optimist. However, in contrast to Gödel, who thought that the mind was somehow distinct from the physical brain and who even gave credence to the idea of an afterlife, Penrose is clear that he regards the mind as purely a product of the brain's physical activity. Since Penrose rejects mechanism, he is led to the conclusion that something non-computable (but non-random) must be going on in the physics underlying the brain. However, all our *current* physical theories would appear

to be basically 'computable' in character (although, as we saw in Chapter 4, it seems surprisingly hard to nail this idea down precisely). So Penrose argues that some radical upheaval in our current understanding of physics will be necessary — a conclusion for which he also finds motivations from within physics itself.

These ideas were elaborated by Penrose in two controversial books: *The Emperor's New Mind* (1990) and *Shadows of the Mind* (1995). Among many other topics, these books contain some fascinating speculations both on what the necessary new physics might be like, and on where in the brain it might be taking effect. However, Penrose freely admits that we have a very long way to go before reaching any detailed understanding of these matters.

Penrose, then, uses Gödel's results to argue that the power of the mind surpasses that of any machine. In this, he goes significantly further than Gödel himself, who claimed that they show only that *either* this is the case *or* there are unanswerable questions of a certain type. What leads Penrose to this stronger claim? Is it simply a predisposition to rational optimism that pushes him into this position? Or are there other arguments pointing in this direction? Let us now turn to a closer examination of Penrose's arguments.[24]

Penrose's case against mechanism

We'll start by reminding ourselves of what would it would mean if the human proof engine *could* be simulated by a machine, say M. As we've seen, we'd then be forced into the other alternative of Gödel's disjunction: there must be absolutely unsolvable mathematical problems. But Gödel's theorem actually tells us more than this: they provide a specific example of a problem that would be unsolvable in this scenario. Namely, they tell us that if the human proof engine were simulated by some mechanism M, then an Arithmetic statement saying 'the mathematical outputs of M are consistent' would be example of such an unsolvable problem. So it's not just that there would exist unsolvable problems — which might, after all, be so obscure and uninteresting that hardly anyone would worry about them. Rather, there would be a very specific unsolvable problem arising in an area of burning concern to us — one that touched on our ability to understand the very workings of our own minds. For if we cannot prove 'M is consistent', then we cannot hope to show (mathematically) why it is that the mechanism of our thinking yields reliable results. To say that there might be *some* questions beyond the reach of our understanding is one thing; to say that this somehow limits our ability to understand *understanding itself* is quite another. Such a conclusion might indeed seem highly unpalatable from the point of view of a basically 'optimist' scientific mindset, and this is surely something to be borne in mind when assessing the plausibility of the mechanist view.

Here is how Gödel put it in the lecture we cited earlier:

> However, if such a rule [i.e. mechanism underlying human mathematics] exists, we with our human understanding could certainly never know it to be such, that is, we could never know with mathematical certainty that all propositions it produces are correct … If it were so, this would mean that the human mind (in the realm of pure mathematics) is equivalent to a finite machine that, however, is unable to understand completely its own functioning.

Regarding the phrase 'understand completely', Gödel added a footnote to clarify that 'the physical working of the thinking mechanism could very well be completely understandable; the insight however, that this particular mechanism must always lead to correct (or only consistent) results would surpass the powers of human reason.'

Later, the philosopher Paul Benacerraf expressed the idea succinctly (though more loosely) as follows: 'If I am a Turing machine, then I am barred by my very nature from obeying Socrates' profound philosophic injunction: *Know Thyself*.'

Penrose himself encapsulates it like this: 'Human mathematicians are not using a knowably sound algorithm to ascertain mathematical truth.' The second chapter of *Shadows of the Mind* offers a detailed defence of this claim against a variety of objections.

One small point of clarification may be helpful here. In the imagined scenario that our mathematical abilities are equivalent to those of a machine M, the point at issue is not whether we can know that we *are* equivalent to M, but merely whether we can know that M, taken on its own terms, is consistent. If, by some miracle, during a walk in the forest we were to stumble upon a description of a machine M that just *happened* to be equivalent to our own mathematical powers, then Gödel's theorem would imply that we couldn't prove 'the mathematical outputs of M are consistent' — regardless of whether we knew, or even suspected, that we were equivalent to M.

Up to this point, things are relatively uncontroversial: thus far, Penrose is in clear agreement with Gödel and many other logicians. But what should we make of the possibility this leaves open: that as mathematicians we are indeed behaving in accordance with some 'unknowably sound' algorithm: that is, with some machine that is consistent for reasons unfathomable to us? Is this merely a ludicrously far-fetched hypothesis — a kind of clutching at straws? Or is there something going for it? Even if such a situation is logically possible, is it *plausible*? Would anyone really want to defend this as a picture of how things were?

Penrose offers essentially three lines of argument aimed at refuting this possibility that our mathematical judgements could be replicated by an

'unknowably sound' algorithm. Let's look at these three arguments in turn, in order of increasing sophistication.

An argument from mathematical experience

Penrose's first point against the 'unknowably sound algorithm' hypothesis is that it appears generally implausible in the light of mathematical experience. After all, when we do mathematical proofs, we *do* understand the the individual steps we're applying, and *can* see (with mathematical certitude) that they are sound! Indeed, the whole basis of our mathematical tradition is precisely the idea that we shouldn't accept any forms of reasoning whose validity we can't 'see for ourselves'.

What's more, Penrose argues, the steps involved in a mathematical argument, if broken down far enough, are sufficiently simple and clear that they can be understood and agreed on by anyone. The canons of correct mathematical reasoning, then, aren't simply locked away in our skulls and encrypted in some impenetrable algorithm, but are public, shared and universally agreed upon. Can one really believe that our common idea of what counts as a mathematical proof arises from some unconscious process so obscure that its very validity can never be known to us? For Penrose himself, 'this is as blatant a *reductio ad absurdum* as we can hope to achieve, short of an actual mathematical proof!'[25]

Of course, one can pick holes in this, for instance by saying that the way things *seem* to us as mathematicians might be quite different from the way things really are at the underlying neurophysiological level. (Penrose knows this.) But we should try to listen to what lies behind the argument. Regardless of whether it's a logical possibility, is it really *plausible* that this 'public domain' phenomenon of our shared criteria for mathematical proof is something that's unexplainable except by recourse to a mysterious algorithm whose working we can never hope to understand?

Let's think through what this strange possibility would entail if it were true. Suppose we grant that for any given step in any proof that we understand, we're able to isolate the reasoning principle we're using and say "Yes, that's a valid principle. We're clearly allowed to use *that* in our proofs." We can do this in turn for any finite set S of reasoning principles, certifying their validity one by one, and conclude that any theorem obtained using just the principles in S would indeed be true: thus, any such S would be a knowably sound proof system. So if every *individual* reasoning principle we might use, and indeed every finite set of such principles, is knowably sound, but our proof mechanism as a whole is *not* knowably sound, then that can only mean that there must be *infinitely* many potential reasoning principles, each of which we'd individually be able to recognize as sound, but such that we'd be

incapable of perceiving the more general fact that *all* of them were. Such an infinite set (on the hypothesis we're considering) might be recognized in a uniform way by some algorithm wired into our brains of which we're not conscious; but if this algorithm were presented explicitly to us, there would be no proof of its soundness *as a whole* that we'd be able to grasp.

This point was affirmed by Gödel, who also made the observation that this wouldn't preclude the possibility of establishing the soundness of the system with 'less-than-mathematical' certainty. To fill in a gap in a quotation we cited earlier:

> … if such a rule exists … we with our human understanding … could never know with mathematical certainty that all propositions it produces are correct; or in other terms, we could perceive to be true only one proposition after the other, for any finite number of them. The assertion, however, that they are all true could at most be known with empirical certainty, on the basis of a sufficient number of instances or by other inductive inferences.[26]

It might seem that this picture only serves to make the imagined scenario appear even more far-fetched. Certainly, Penrose finds it quite implausible:

> Of course, none of this tells us that such [a system] is an impossibility; but from the point of view of our mathematical understandings, its existence seems very unlikely. In any case, there is, at present, not the slightest suggestion as to the nature of a plausible such [system], nor is there any hint whatsoever of its existence.[27]

Is this a fair assessment? Or can one, in fact, detect suggestions or hints anywhere of the existence of such an 'unknowably sound' system? Our earlier discussion of reflection might already have provided a glimmering of something like this, and we shall be keeping our eyes open for further clues as we proceed.

An argument from evolution

Penrose's second argument has to do with origins. *If* there is some unknowably sound algorithm lurking in our brains, how could it possibly have got there? An ability to do sophisticated mathematical reasoning, says Penrose, would hardly have been of survival value in the environment in which our ancestors evolved. (In fact, it might even have been a *disadvantage*, because anyone who took to mulling too deeply over mathematical problems would have been less likely to notice the sabre-toothed tiger coming up behind.) So why on earth should nature have gone to the trouble of equipping

us with a sophisticated algorithm for generating infinitely many reasoning principles — let alone an algorithm so deviously subtle that its very correctness lies beyond our comprehension — when just a tiny handful of these principles would have sufficed for day-to-day survival? Penrose's own view is that it is exceedingly unlikely that such an algorithm could have been generated purely by Darwinian evolution.

One might well feel that there is at least some kind of puzzle here. Most specifically, our ability to grasp and reason about the concept of *the infinite* might seem particularly baffling, given that a such a concept has no counterpart within the world of our sensory experience. This does indeed seem to call for an explanation, and it may be that we are a long way away from having one.

At the same time, though, I think many would say that our current inability to explain this is very weak as an argument that it *didn't* happen. Isn't this simply an example of what Richard Dawkins calls an 'argument from personal incredulity'?[28] In a response to Penrose, Hilary Putnam points out that 'just this objection has been made to almost every evolutionary explanation … In case after case, an answer has been found … but the answer typically involves generations of researchers'.[29]

Of course, it may be that only by *trying* to find an explanation will we discover whether there really is some deeper problem here. Progress in this area will most likely need to await a much fuller scientific understanding of the brain mechanisms underpinning mathematical thought than we can lay claim to at present.

I also can't resist chipping in on another possible weak link in the argument at this point. This concerns the difference between saying that our *observable* mathematical behaviour (i.e. the set of theorems we can certify) coincides with that produced by an unknowably sound algorithm, and saying that some such algorithm is actually *enacted* within our physical brains somehow (whatever that means). Perhaps the former is true, but not the latter? For example, Penrose himself seems to suggest that our minds have some as yet unexplained faculty for 'perceiving' a realm of objective mathematical truth. If this were so, then to be sure, the richness and complexity of the mathematics that we perceive could perfectly well reside in this objective realm itself rather than in our brain structures; and this would get round the evolutionary difficulty if there is one. But on this scenario, what would exclude the possibility that the underlying (new) physics is perfectly computable in character? Perhaps this physics only enables us to see a certain 'distance' into the mathematical realm, and the soundness of our own understanding lies beyond the reach of our mathematical 'vision'? In this case, the truths we could see might well coincide with those generated by some conventional computer program, though the computer wouldn't be 'seeing' their truth in anything like the way we do.

Perhaps Penrose wouldn't be so strongly opposed to this (wildly speculative) hypothesis. After all, his main goal is to argue that some radically new kind of physics is needed to explain human understanding, and his arguments via Gödel's theorem and non-computability are really just a stepping-stone towards this conclusion. However, in the light of the above reflections, one might wonder whether this stepping-stone is something of a red herring. Perhaps there's actually a stronger case for the main claim ('something radically new is needed to explain human understanding') than there is for any kind of non-computability in physics? We shall be picking up this idea again in later chapters.

Penrose's 'new argument'

We now come to the third of Penrose's arguments: a fascinating, subtle and mind-bending piece of reasoning which (in my opinion) is still much less well known than it deserves. The argument first appeared in *Shadows of the Mind* (in a somewhat complex and elaborate form), as an attempt at rigorously refuting the possibility of an 'unknowably sound' algorithm underlying our thinking. Later, Penrose distilled the essence of the argument into a much simpler form, and it's this version that I shall discuss here (with minor differences in presentation). Admittedly, this will be a bit of a digression from our main line of thought, so you might prefer to skim rather lightly over this section. On the other hand, if you have the patience to work through this intriguing argument line by line, you may well find it rewarding.

The idea is an ingenious one, and even one of the leading critics of the argument (the logician Peter Koellner) calls it 'subtle and sophisticated' and 'a brilliant piece of reasoning'.[30] In contrast to the previous two arguments — both clearly informal in nature — Penrose's so-called 'new argument' is an ambitious attempt at a rigorous logical demonstration that, for *any* formal system *F* of axioms and rules that we might stumble upon in the forest, the hypothesis that *F* matches our own mathematically certain knowledge leads to a contradiction! (I shall follow Penrose here in phrasing the argument in terms of formal systems rather than machines, though it makes no essential difference for these purposes.) Since it doesn't matter here whether *F* is knowably sound or not, this argument, if valid, would indeed close off the loophole that we've been discussing.

The key idea is this. Suppose we're given some formal system *F*. Then even if we don't know whether *F is* the system underpinning our mathematical knowledge, we can certainly contemplate the possibility that it *might* be, and we can ask what would follow from this if it were true. In other words, we can consider the class of all statements P such that we know (with mathematical certainty) that *if we are F*, *then* P. So let us now speculatively adopt the

hypothesis that *we are F* (we understand this phrase just as an abbreviation for 'the theorems provable in *F* are exactly the theorems we can establish with mathematical certainty').

What would follow from assuming this? Firstly, since everything knowable with certainty must be true, it would follow that everything provable in *F* was true. But secondly, since the statement 'we are *F*' would also be true, we could extend *F* by adjoining the extra axiom 'we are *F*', and still have a truthful theory. Let's denote this new system *F* + 'we are *F*' by F^+. The theorems provable in F^+ will be exactly the statements that we can see to be consequences of 'we are *F*'.

So we've shown (always under the assumption that we are *F*) that F^+ is truthful, which implies that F^+ is consistent. By the second incompleteness theorem, this is something we couldn't hope to prove within F^+ itself.

But now here's a funny thing. *We* have just proved that if we are *F*, then F^+ is consistent. But implications of this kind ('if we are *F*, then such-and-such') are exactly what F^+ was designed to show! So since *we* have seen that 'F^+ is consistent' is a consequence of 'we are *F*', it must be that 'F^+ is consistent' is provable in F^+ after all. We now have a contradiction.

Before reading further, you might well like to stop and ponder this argument for yourself to see what you make of it.

* * * * *

Speaking for myself, the first time I grasped the argument, it seemed that there was even something quite spooky about it. Certainly, there are things here that might make one feel uneasy: the subtly self-referential twists, and in particular the attempt to reason about our own powers of reasoning. But does any of this actually invalidate the argument? Which steps of the argument, if any, are open to challenge? Several logicians and philosophers have undertaken to examine the argument closely, and I shall draw on some of their ideas in what follows.[31]

First, one might feel that the argumentation can't lay claim to 'mathematical certainty' in the sense that's required, because it makes reference to the concept of 'establishing something with mathematical certainty' — which is not itself a concept we've made mathematically precise. On the other hand, we don't actually need to *define* this concept in order to conduct the argument: all we require are some basic *properties* of this notion which can be easily formulated. Indeed, Koellner has shown how a version of Penrose's argument can be conducted perfectly formally with the help of a *truth* predicate **T** (broadly similar to the one we considered earlier) and a *knowability* predicate **K**, subject to some simple and clearly stated laws. In this way, it seems, the reasoning can indeed be made sharp enough to make at least a bid for 'mathematical certitude'.

One might also feel nervous about a principle that's invoked early on in the argument: the idea that *everything we know with certainty is true*. Of course, we all *believe* this — to deny it would seem tantamount to an abnegation of faith in rationality itself! But is it right to say that we *know* this with certainty — with the same kind of certainty to which the statement itself alludes? For myself, I confess to having some discomfort about this. Nevertheless, it does seem to me that this is a position one *could* adopt, given a certain kind of 'rationalist' faith in a concept of idealized knowability — so it's of interest to see what such a principle implies. Indeed, it would seem already quite remarkable if such a position, in and of itself, brought one into conflict with the mechanist thesis that 'we are (some formal system) *F*'. So let's agree to accept this principle for the sake of argument, to see where it leads.

This brings us to what might be the deepest source of unease regarding Penrose's argument: namely, the 'self-referential' way in which the concepts of truth and knowability are applied. Indeed, Koellner's analysis shows that for the argument to work, the truth predicate **T** needs to be applicable to statements that themselves involve **T** (and likewise **K** needs to be applicable to statements involving **K)**. But we've already seen how a system that can talk about *its own* concept of truth is liable to inconsistency, by Tarski's argument. So this raises the suspicion that Penrose's argument depends crucially on some combination of reasoning principles which is in fact subtly paradoxical.

Of course, we shouldn't let it rest there. If some combination of individually plausible principles leads to a paradox, we should try to understand what's at the root of the paradox in order to see which of our principles ought to be modified, and how. In the present case, it turns out that there are several ways in which one can do this, with no general agreement as to which is best. Perhaps one of these modifications might lead to a system that's free from paradox but still suffices for Penrose's argument?

This question has been taken up by Peter Koellner, who has looked closely at one of the most promising candidates for such a system. Via some impressive technical analysis, he shows that not only does this system fail to support the specific argument offered by Penrose, but that *no* proof within this system of the non-mechanist conclusion 'we are not *F*' is possible. Of course, it could be that some *other* modification of our original system would let one prove this, but in the light of Koellner's work, this seems quite unlikely. Koellner therefore leaves his opponent with the challenge of coming up with a suitable system. And this seems to be about where this particular debate has currently got to.

But what's knowable truth *actually* like?

Our whole discussion of this possibility of an 'unknowably sound' algorithm that matches our capacity for mathematical understanding has by now drifted into quite speculative waters: *could* the extent of knowable mathematical truths, conceivably or in theory, be like this or that or the other? So I'd now like to propose a different tack. Rather than simply speculating on what the extent of knowable mathematical truth *might* be like, I would suggest that it's possible to take a closer look and learn something more about what it actually *is* like! Now that our philosophical discussions have given us a fair idea of what the questions are, we can see whether a more in-depth study of the actual nature of mathematics has any further light to shed on them.

Insofar as there's anything distinctive about the particular approach to these questions that I'm advocating in this book, this is it. I hope to persuade you that the accumulated discoveries of mathematical logic from Gödel onwards have actually given us quite a detailed and compelling picture of the nature of 'knowable mathematical truth' and how it is structured. Once this picture is in place (I claim), the questions we've been asking will appear in a rather different perspective, and we will be rewarded with a much more satisfying understanding of the whole situation.

We've already made a good start in building up this picture through our discussion of Gödel's theorems and the idea of reflection. But to make further progress, we will now need to resume our quest and venture into some more advanced areas of mathematical logic. So let's now rejoin Anna and Emily on their grand ascent, and try to get to grips with some of the remarkable discoveries they made towards the end of the story.

Staircase induction and well-foundedness

Remember that the girls started from Peano Arithmetic — a system they could clearly see to be truthful — and then embarked on a relentless process of applying reflection over and over again, climbing up to ever more powerful systems which they could also see to be truthful. But how far can this process be carried? If we just obsessively keep on trying to climb the ordinal staircase as far as we can, what happens? Is there some kind of a limit to how far we can travel? And if so, why?

We've already followed the girls in their ascent through the levels ω, ω^{ω}, $\omega^{\omega^{\omega}}$, $\omega^{\omega^{\omega^{\omega}}}$, …, at the limit of which they arrived at the moderately mind-mangling ordinal known as ε_0. But here, something new and unexpected happens. Broadly speaking, this has to do with the question: How do we know that ε_0 really *is* an ordinal? This is something we need to be assured of if we are to trust the system PA_{ε_0}. Let's explain all of this more carefully.

Just suppose that one day you find yourself on some weird infinite staircase, which may or may not be the same as the ordinal staircase the girls are on. On each level is a minter that generates some class of axioms. Let's say a level is *sound* if all the axioms generated by the minter on that level are meaningful and true. To begin with, you don't know that *any* of the levels are sound; but let's suppose that you do know the following.

Staircase Induction Principle:
For any level α, *if* all the levels below α are sound, *then* α itself is sound.

You yourself are sitting at some level, say β. What would it then take to convince you that β was sound?

Let's first consider a situation where you *wouldn't* be able to do this. Suppose that your staircase had no 'bottom level'. For instance, suppose you are sitting on level 5, and below you are not only the levels 4, 3, 2, 1 and 0, but continuing below them, an infinite descending sequence of levels -1, -2, -3, …. Then there'd be nothing in the Staircase Induction Principle to rule out the possibility that *all* levels are unsound. In other words, the structure of this particular staircase doesn't lend itself well to 'justifying' the soundness of individual levels.

But now suppose that level 0 *were* the bottom level (as is the case in the real ordinal staircase). You are again on level 5, looking down on levels 4, 3, 2, 1, and 0. In this case, things would be quite different. You would then know, for starters, that level 0 was sound. (This is because there'd be *no* levels below 0, so it would be trivially true that all levels lower than 0 were sound, and Staircase Induction would then tell you that 0 was sound.) And once you knew that 0 was sound, you could use Staircase Induction to tell you that 1 was also sound, and then that 2 was sound, … and so on up to level 5. So such a staircase would allow us to 'justify' at least some of its levels (we haven't said anything about what happens above level 5).

We can recast the idea in the following way, which is closer to how Anna described it in the story. You are trying to show that level β is sound. You might try to do this by supposing β *wasn't* sound, and trying to derive a contradiction from this. So let's see how we get on. If β wasn't sound, then by turning the Staircase Induction Principle back to front, there would have to be some level below β — call it β_1 — that was also unsound. But then, for exactly the same reason, there'd also have to be another unsound level somewhere below β_1 — say β_2. In this way, we might obtain a whole descending sequence of unsound levels:

$$\beta > \beta_1 > \beta_2 > \beta_3 > \ldots$$

Does this give the contradiction we're hoping for? Not as it stands. We've already seen that, depending on the staircase, there *might* be an infinite descending sequence of levels, and they might all be unsound. But now suppose that level β of the staircase has the following special feature.

Well-foundedness Property:
There is no infinite descending sequence starting at β.

In this case, our sequence β, β_1, β_2, β_3, … will eventually reach a level with nothing below it, and we will have shown that this bottom level is unsound. But this *does* give us a contradiction, because as we saw above, it's easy to show that a bottom level (if it exists) is sound! So if we can assure ourselves that β has the Well-foundedness Property, we can complete our proof that level β is sound.

Now back to Anna and Emily. They are, in fact, exactly in the situation of being on a staircase and knowing that the Staircase Induction Principle holds, essentially for reasons we've already explained.[32] So if they have reached some level β and want to convince themselves that it's sound, their problem is precisely to show that β *is well-founded*.

This may seem all well and good, but actually, we encounter a serious obstacle at this point. The difficulty is that we have no precise way of even stating the property 'β *is well-founded*' using the means currently at our disposal. Remember that we (and the girls) are trying to see how far we can get simply on the basis of our understanding of Whole numbers, so any other concepts we wish to refer to will need to be 'grounded' in the concept of Whole numbers in some way. However, the definition of well-foundedness makes reference to a quite different kind of concept: that of an arbitrary *infinite descending sequence* of ordinals. Even if we're wishing to say that there's *no* such thing, we at least need to be able to express this concept! But this is a problem, because we don't know any way to reduce the general concept of an 'infinite sequence' to sometime couched in terms of Whole numbers. (We shall explain all of this much more fully in the next chapter.)

On the other hand, we can certainly talk about lots of *examples* of potentially infinite sequences, and for any particular sequence *s* that we're interested in, we can state the property: 'if *s* is a descending sequence of ordinals starting at β, then *s* eventually bottoms out at 0'. So strictly speaking, Well-foundedness for β will have to be expressed by an *infinite* family of statements, one for each sequence we might be interested in — much as within Peano Arithmetic itself, the Principle of Induction was captured by an infinite family of axioms generated by the same minter. This may seem like a rather abstruse technical point, but it will come back to haunt us later.

Subject to this caveat, we can now ask *which* levels of the staircase the girls can prove to be well-founded. It turns out that within PA, it's possible to prove

that ω is well-founded, and that ω^ω is well-founded, and that ω^{ω^ω} is well-founded, and so on. In fact, for any level α below ε_0, *considered individually*, it's possible to prove in PA that α is well-founded. But it also turns out that it's not possible in PA to prove the single general statement: '*All* levels below ε_0 are well-founded'. (Does this remind you of anything?) And without this general statement, we can't conclude that ε_0 itself is well-founded. So the first hint of a new kind of difficulty begins to assert itself at level ε_0.

However, the journey doesn't end here. We are only saying that we can't prove *within our base system* PA that ε_0 is well-founded. But we long ago convinced ourselves that the stronger system PA_1 was truthful, so a proof of well-foundedness *within* PA_1 would be equally legitimate. And it turns out that PA_1 can indeed prove 'ε_0 is well-founded'; and this allows us to continue our upward journey.

As a brief aside, it's worth noting that knowing 'ε_0 is well-founded' is exactly what we need to analyse the hydra battle we discussed earlier. Roughly speaking, for any given hydra h, it's possible to define an associated ordinal α_h less than ε_0, in such a way that the hydra's ordinal always *decreases* with every round of the battle — even if the hydra appears to get larger.[33] But we now know that *any* decreasing sequence of ordinals below ε_0 eventually crashes out at 0 — so any hydra battle eventually terminates, and Hercules wins! This is essentially why PA_1 suffices for solving the hydra problem, and it also gives a hint of why PA failed to do so.

Ordinal bootstrapping

Now that we've decided to use PA_1 for our well-foundedness proofs, how much further can we get? A very, very long way indeed.[34] But even so, the power of PA_1 will eventually give out, and we'll be forced to step up to PA_2 in order to continue further. And so it goes on: the stronger the 'base system' we use, the more levels of the staircase we'll be able to establish as well-founded. This was the point illustrated by our tortoise, whose plodding progress up the ladder of theories PA_0, PA_1, PA_2, … enabled the girls to continue their dizzying elevator ascent into ever wilder regions of infinity. The further the tortoise climbed, the more of the staircase was opened up for exploration.

This leads us to conceive the following megalomaniacal scheme for constructing an increasing sequence of ordinals α_0, α_1, α_2, …. First, we set $\alpha_0 = 0$, and note that on the basis of trust in PA_0, we can obtain a justification for PA_1 by means of reflection. Next, we look for the first level which can be shown well-founded in PA_1 but not in PA_0. This turns out to be ε_0, so let's set $\alpha_1 = \varepsilon_0$. We now know that PA_{α_1} is sound, so by reflection, PA_{α_1+1} is also sound. So we can now ask how far we can get when the tortoise reaches α_1: or more precisely, what is the first level that can be shown well-founded in PA_{α_1+1} but

not in PA_{α_1}. This will be such an unthinkably vast ordinal that I shall make no attempt to describe it; but anyway, let's call it α_2. So we now have a justification for PA_{α_2}, and hence for PA_{α_2+1}.

Continuing in this way, we get an outrageously accelerating sequence of levels α_0, α_1, α_2, α_3, … rising to ever more ridiculous heights, and an attendant sequence PA_{α_0}, PA_{α_1}, PA_{α_2}, PA_{α_3}, … of ever more powerful axiom systems. Starting from a trust that PA_{α_0} = PA is truthful, we can bootstrap ourselves up to a trust in PA_{α_1}, and from there to a trust in PA_{α_2}, and so on, since each of these systems (with some help from reflection) provides the means we need to reach up to the next. Such a sequence is known technically as an *autonomous progression* of axiom systems.[35]

The limit of predicative mathematics?

Part of our purpose in this chapter has been to sound out the limit of what's known as *predicative* mathematics. (Note that 'predicative' is a different word from 'predictive'!) We shall be explaining and qualifying the meaning of this term much more thoroughly in the next chapter — but for the time being, we may broadly identify predicative mathematics as that part of mathematics that becomes accessible to us when we accept the basic concept of the Whole numbers (including their surveyability), even if we don't accept any further fundamental concepts beyond this.

As we've seen, this 'predicative' style of reasoning — reasoning grounded in our understanding of Whole numbers — can carry us a very long way indeed, and Peano Arithmetic is (literally) just the first step. By relentlessly applying reflection to work our way up the staircase of ever stronger systems, and then, each time the staircase itself seems to be in danger of giving out, using our bootstrapping trick to open up a bit more of it, we've been able to scale some stupendous heights sufficient to boggle even the wildest imaginations, as represented by our ultra-increasing sequence of levels α_0, α_1, α_2, α_3, ….

But can we go any further than this? Does this ultra-increasing sequence *itself* have a limit — some ultra-landing from which we can look down on everything we've talked about so far?

It's at just this point that we seem to reach some kind of ceiling. The generally accepted view among logicians today is that if we are indeed asking about how far we can get with just predicative reasoning — that is, with just our understanding of Whole numbers — then there is indeed a fundamental obstacle to passing to the limit of our ultra-sequence. On the one hand, we can certainly define and talk about this supposed limit level — it's technically known as Γ_0 (Gamma-zero) — but there is a genuine difficulty involved in proving that it is well-founded, which is what we need in order to be assured

that the corresponding system PA_{Γ_0} is truthful. Put briefly, any attempt to prove that Γ_0 is well-founded turns out to require a system at least as strong as PA_{Γ_0} itself — the very system we are trying to justify.

This vertiginous ultra-limit Γ_0 is, of course, the precarious and ghostly platform on which we left our heroines stranded at the end of the current episode. It was first identified as the 'outer limit of predicative reasoning' through the work of Solomon Feferman and Kurt Schütte, who (independently) mapped out a version of our epic ascent via a magnificent mathematical *tour de force* in the early 1960s.[36]

The claim that PA_{Γ_0} *is* the ceiling of where we can get to with purely predicative reasoning (in the sense in which we're using the term here) should perhaps be seen as a kind of philosophical thesis rather than as a precise mathematical statement amenable to proof — a claim comparable in status to the Church-Turing thesis for 'mechanically computable' functions. But the consensus of the experts seems to be that the case for this 'Γ_0 thesis' is a pretty strong one.[37] Not only has it withstood 50 years of scrutiny, but it has been somewhat strengthened by the discovery that other approaches to determining the 'limit of predicativity' also converge on this same level Γ_0.

What isn't so easy to give a good intuition for *why* there's a fundamental problem in passing to the limit of our ultra-sequence $\alpha_0, \alpha_1, \alpha_2, \alpha_3, \ldots$. Indeed, if you've managed to follow the discussion so far, it's quite possible that you might be reluctant to accept this. Surely (I hear you cry), once we know how to get from α_0 to α_1, and from α_1 to α_2, and from α_2 to α_3, we can just 'step outside' this whole process one more time, and notice that the same procedure is being applied at every stage? Wouldn't we then be able to conclude that for *all* n, the level α_n was well-founded — from which it would readily follow that Γ_0 itself was well-founded?

This may sound intensely plausible. Certainly, I found myself thinking along lines like this for quite some time when I was getting to grips with these ideas. But let me make a stab at explaining what the problem is.

It turns out that there's a real difficulty in suitably formulating and proving the *general* argument that gets us from each α_n up to α_{n+1}. Granted, one could spot that there was some general pattern at work in the passage from α_0 to α_1, from α_1 to α_2, and from α_2 to α_3; and this might indeed make it seem highly plausible that the same thing will work for any n. But how might we rigorously formulate the general principle we use to climb from α_n up to α_{n+1}?

One might, for example, try phrasing it like this:

For all n, if α_n is well-founded, then α_{n+1} is well-founded.

Unfortunately, though, this isn't something we can even meaningfully state at present. A few pages back, we noted that 'α is well-founded' isn't, strictly speaking, an idea that it's possible to capture by a single formula within our

current framework. Rather, it's a rather loose shorthand for a potentially infinite family of formulae. But if we don't have a *single* formula to express well-foundedness, we can't even properly formulate the statement we've displayed above, let alone prove it.

As a second try, we might try to express the key induction step like this:

For all n, if PA_{α_n} is truthful, then $\mathrm{PA}_{\alpha_{n+1}}$ is truthful.

Can't one formulate a statement like this, and use it to establish the general truth that *all* PA_{α_n} are truthful, and hence that PA_{Γ_0} is truthful? Indeed one can. But what means would be necessary to conduct such an argument? Even to express the general property 'PA_{α_n} is truthful' would require some kind of **T** predicate that could talk about the truth of any formula of any PA_{α_n}. In other words, it would basically have to be the **T** predicate at level Γ_0 itself, so we would in effect require the power of PA_{Γ_0+1} to carry out the argument. (This, by the way, is why the tortoise was found to be just one level above Anna and Emily in the story.) But of course this won't do as a justification for PA_{Γ_0}, because it presupposes that we believe already that PA_{Γ_0+1} is trustworthy.

One can go on like this, trying to devise other clever ways to get round the problem; but experience shows that in each case, the proposed argument will either fail to be expressible, or else will subtly presuppose a trust in something at least as powerful as PA_{Γ_0} (or equivalently the well-foundedness of PA_{Γ_0}). Eventually (if my own experience is anything to go by), one's resistance will be worn down and one will be led to concede that the 'Γ_0 thesis' is almost certainly correct.

There is a delicate problem here. I realize that a few of my statements earlier on might have given the impression that *I myself* somehow have secret inside knowledge that all of the systems PA_{α_n} *are* indeed truthful, despite my claims that we can't actually prove this. However, this is merely a case of a somewhat loose manner of speaking. Strictly, the general claim that all the α_n are well-founded ought, from our present perspective, to be regarded as no more than a highly plausible conjecture. Whether it appears as more than this from some *other* perspective, we shall discover in the next chapter.

An 'unknowably sound' system?

Let's now step back and take stock of what we've discovered. If the picture we've outlined is correct, then the system PA_{Γ_0} looks very suspiciously like something we touched on earlier as a mere hypothetical possibility: an infinite, computably generated system of axioms which can be perceived to be true *individually*, but not *collectively*. That is, any given axiom of PA_{Γ_0} will

occur already at some level PA_{α_n} and can be seen to be true, but the general fact that all of them are true appears to be beyond our ken.

Of course, this isn't *exactly* what we were talking about earlier, because here we're only considered what can be known via so-called 'predicative' reasoning, and we haven't yet seen what other kinds of mathematically acceptable reasoning might be on offer. Nonetheless, what we have here is certainly close in spirit to the situation we hypothesized earlier, and it serves as an interesting specimen of a phenomenon that really can arise. The essential point is none of our now familiar tricks of reflection or Gödelization or 'stepping outside the system' are of any help in allowing us to proceed further. Our 'understanding' of the content of PA has indeed turned out to be remarkably fertile, and has carried us way beyond PA itself, but eventually, at stage Γ_0, its possibilities are exhausted. So if there is any way of 'seeing' that PA_{Γ_0} and even stronger systems are truthful, this will have to be via some completely new idea — something that takes us beyond the limits of predicative mathematics.

What we are saying, then, is that *for all Anna and Emily know at this stage*, it appears that PA_{Γ_0} *could* have the status of an 'unknowably sound' system equivalent to their own truth perceptions. (Of course, it would be ridiculous to suggest that the whole monstrous definition of PA_{Γ_0} is somehow wired into our brains; however, it turns out that there are other, much simpler and cleaner systems that prove exactly the same Arithmetical truths as PA_{Γ_0}.[38]) So this whole possibility of an unknowably sound system matching our mathematical perceptions might now seem not quite so far-fetched as it did at first.

Actually, this situation isn't too dissimilar to something we've seen before. In the last chapter, we met the idea that the means of proof acceptable to a mathematician of Hilbert's 'finitist' school are precisely captured by the system known as PRA. Now, PRA of course can't prove its own truthfulness, or even consistency, so the finitist can't 'see' that PRA as a whole is truthful. However, for any particular axiom of PRA, or indeed for any finite number of them, the finitist *can* see that everything provable from those axioms is true. So for the finitist, PRA plays the role of the 'unknowably sound' system matching his truth perceptions.

How do we get beyond PRA? For our finitist, there is simply no way to do so: indeed, any attempt to 'justify' PRA as a whole would itself require a system stronger than PRA. As we saw in the last chapter, there is a fundamental Gap between the world of PRA and a system like PA; and the only way to bridge this gap is to take on board some completely new philosophical principle: in this case, the surveyability of $\mathbb{N}$, which takes us out of the world of finitism and into the wider world of *predicativist* mathematics. Once we've made this jump, we can readily accept the system PA; and from this standpoint, to be sure, we can look down on the weaker system PRA and see that it is truthful.

In essence, what we've seen in this chapter is basically a repetition of this same pattern, only on a much grander scale. For our supposed predicativist mathematician — one who accepts the surveyability of $\mathbb{N}$, but nothing beyond this — we've argued that our system PA_{Γ_0} plays the role of the unknowably sound system behind her truth perceptions. As we've seen, there doesn't seem to be any way to 'break the Γ_0 barrier' via any predicatively acceptable kinds of reasoning.

So we're now left with an obvious question. Is there some further philosophical principle or belief which, should we be willing to accept it, enables us to bridge the Gap between the world of predicativist mathematics and some still higher vantage point? What philosophical issues would this new belief raise, and what might this further vantage point look like? To learn the answers, we must now rejoin our heroines on their airy perch to discover what it is that Anna has just glimpsed through the clouds.

[1] Actually, it *is* possible for a system to express its own truth predicate if we're willing to sacrifice something else (for example, if we abandon classical two-valued logic). For a discussion of some of the trade-offs and possible choices, see H. Leitgeb, *What theories of truth should be like (but cannot be)*, Philosophy Compass, 2(2), 276,-290 (2007).

[2] The argument is actually a fairly easy adaptation of Gödel's original construction of a glyph statement that says, in effect, 'I am not provable', which we outlined in Chapter 6, endnote [?]. All we have to do is to replace 'provable' by 'true', using the hypothesized formula T to express the idea of truth in glyph language. Once we have a self-denying glyph statement of this kind, a contradiction follows easily.

[3] Curiously, Epimenides' statement found its way into the Bible (Titus 1:12-13), where the apostle Paul adds the comment: 'This witness is true'. I can't help wondering whether the joke was deliberate.

[4] Actually, it's not entirely trivial to do all this, and there's some detail I'm skating over here. What's perhaps not too hard to believe is that all the relevant operations on numerical codes can all be performed by tube machines: for instance, there's a tube machine which, given codes for formulae P and Q, computes a code for P & Q; and a tube machine which, given a code for a statement $t = u$ (for instance, $5 + 7 = 3 \times 4$), computes the values of t and u and tests whether they are indeed equal. And we saw in Chapter 4 that statements about the results tube machine computations can (with a bit of work) be expressed in the Arithmetic. These ideas give us basically all we need to formalize the intended axioms for **T**.

[5] Throughout this chapter, my treatment of the reflection hierarchy based on **T** predicates closely follows Halbach (2010), to which the reader is referred for the full story (see especially Chapters 9 and 22).

[6] See (Gödel 1995), page 309. The lecture in question was the so-called 'Gibbs Lecture' which Gödel gave to the American Mathematical Society in December 1951: *Some basic theorems on the foundations of mathematics and their implications*. Although Gödel seemingly intended to publish the text of the lecture, he never did so, and it was not until 1995, with the posthumous publication of his collected works, that this astonishing gem became readily accessible to the world at large. In the light of this, we can now see how remarkably far-sighted Gödel was. As Peter Koellner puts it, 'He in effect anticipated every move in the subsequent debate and he was able to avoid the pitfalls that beset others.' (In P. Koellner, *On the question of whether the mind can be mechanized, I: From Gödel to Penrose*, Journal of Philosophy 115(7) (2018), page 359.)

[7] See A.M. Turing, *Systems of logic basic on ordinals*, Proc. London Mathematical Society 45(3), 161-228 (1939). Reprinted in Davis (1965).

[8] I shan't give full details here of how the axioms for $\mathbf{T}_\alpha$ are formulated in general: see Halbach (2010), Chapter 9.

[9] There are some subtleties I'm glossing over here. Although I'm cheerfully writing PA_α for 'the extension of PA reached at level α', strictly speaking the definition of this system depends not only on α, but on the particular way in which α is represented or 'notated' (see the next section in the main text). This matters because different notations for the same ordinal can in principle give rise to quite different extensions of PA (this was already noted by Turing in his 1939 paper). For our present purposes, however, this won't matter too much, because for all the ordinals we'll be considering, everything works fine if we stick to certain recognizably 'standard' kinds of codings. (For technical details, see for example Chapter 7 of Pohlers (2009)).

[10] See Gödel (1995), page 310.

[11] Indeed, Peter Koellner has shown via a thorough and careful analysis that if the background assumptions regarding these concepts are made fully explicit, then the disjunction has the status of a provable mathematical theorem. See P. Koellner, *Gödel's dichotomy,* in Horsten and Welch (2016).

[12] I have been stimulated by and benefited from the writings of (among others) George Boolos, David Chalmers, Solomon Feferman, Peter Koellner, Per Lindström, Roger Penrose, Hilary Putnam and Stewart Shapiro on these topics. In some places, my discussion here responds to points raised by these authors.

[13] Gödel alludes in passing to the implicational form in his Gibbs lecture: see Gödel (1995), page 310, footnote 15.

[14] To make this precise, one could define the *size* of a tube machine M to be (for instance) the number of digits in its decimal code, and the size of a glyph formula to be simply the number of symbols it contains. In this case, if M is a machine of size m, then we could express 'M is consistent' by a glyph formula of size $m+c$, where c is some fixed constant which we could explicitly calculate if we wanted (basically the number of symbols it takes to express ' is consistent'). So a tighter version of the implication might read: For any m, if H can be simulated by a machine of size m, then there are Arithmetic statements of size at most $m+c$ which H can't resolve.

[15] If you happen to warm to this particular kind of wrangling, you can find plenty more of it in Penrose's book *Shadows of the Mind* (Penrose (1995)).

[16] The lecture, on the subject of the Automatic Computing Engine (ACE), was given to the London Mathematical Society on 20 February 1947. See Turing (1992), pp. 106-7; also Hodges (2014), Chapter 6.

[17] One important question that arises is whether some kind of upper bound is put on the length of the review process. It's particularly interesting to explore what happens when no such bound is imposed: in this case, we may say a theorem is *known-in-the-limit* if it is proposed by the community and *never* thereafter retracted under any possible behaviour of the reviewer. (For a more detailed outworking of one version of this idea, see S. Shapiro, *Incompleteness, mechanism, and optimism*, in Bulletin of Symbolic Logic 4(3) (1998).) In this scenario, one can formulate a variant of Gödel's implication along the following lines: for any computably constructed community of mechanical mathematicians, there are true Arithmetic statements that are not *knowable-in-the-limit* by this community. (In this situation, the Arithmetic statements in question will not be Predictive ones, but will live at the next level up in the so-called *arithmetical hierarchy*.)

[18] See S. Feferman, *Are there absolutely unsolvable problems? Gödel's dichotomy*, in Philosophia Mathematica III (2006), page 11.

[19] See Hilbert (1970), pp. 378-87. On Hilbert's death in 1943, the concluding words of this lecture were inscribed on his tombstone: *Wir müssen wissen. Wir werden wissen.*

[20] For instance, in his 1950 essay *Can a machine think?*, Turing shows a curious openness to the possibility of Extra-Sensory Perception. And a BBC radio talk he gave in 1951 hinted at the possibility that quantum indeterminacy might have some role to play in explaining 'free will' (see Hodges (2014), pp. 441-442).

[21] See S. Feferman, *Are there absolutely unsolvable problems? Gödel's dichotomy*, Philosophia Mathematica III, (2006), pages 15-16, and S. Feferman, *Penrose's Gödelian argument*, Psyche 2 (1996), pages 21-22.

[22] See P. Koellner, *On the question of whether the mind can be mechanized, II, Penrose's new argument*, Journal of Philosophy 115(9), 2018, page 480. A similar possibility had earlier been floated by Per Lindström.

[23] Both quotations are from R. Penrose, *Beyond the doubting of a shadow*, Psyche 2(23) (1996), Section 4.

[24] Arguments in a similar vein had previously been advanced in Nagel and Newman (1958), and at greater length by the philosopher John Lucas in various writings from 1961 onwards. For definiteness, I shall concentrate here

on Penrose's versions as being easily the most developed arguments of this kind.

[25] See *The Emperor's New Mind* (1999 edition), Chapter 10, pp. 539-540.

[26] See (Gödel 1995), page 309. In a footnote, Gödel elaborates on how, with suitable advances in brain physiology, one might come to know with 'empirical certainty' that the system was sound.

[27] See *Shadows of the Mind*, Section 3.3, especially pages 135-137. The quotation is from page 137.

[28] See R. Dawkins, *The Blind Watchmaker*, Penguin (2006), page 38.

[29] See H. Putnam, *The Gödel theorem and human nature*, in Baaz *et al* (2011), page 330.

[30] In P. Koellner, *On the question of whether the mind can be mechanized, II: Penrose's new argument*, Journal of Philosophy 115(9) (2018).

[31] In addition to Koellner's paper cited above, see D. Chalmers, *Minds, machines and mathematics: A review of 'Shadows of the Mind'*, Psyche 2, 11-20 (1995); Per Lindström, Remarks on Penrose's 'new argument', Journal of Philosophical Logic 35, 231-237 (2005); and S. Shapiro, *Mechanism, truth, and Penrose's new argument*, Journal of Philosophical Logic 32, 19-42 (2003).

[32] Quick technical recap: The Staircase Induction Principle is true at $\alpha = 0$ because PA is truthful; it's true if α is some successor ordinal $\gamma+1$ because $PA_{\gamma+1}$ is clearly truthful if PA_γ is; and it's true if α is the limit of an increasing sequence $\alpha_0, \alpha_1, \alpha_2, \ldots$ because PA_α merely collects together all the axioms from PA_{α_0}, PA_{α_1}, PA_{α_2}, …. And by definition, these are the only three ways of manufacturing computable ordinals.

[33] It's worth seeing into a bit more detail the way we assign ordinals to hydras. Given a hydra, we label each of its heads and nodes with an ordinal, starting with the heads and working towards the root. First, all the heads get the label 0. Next, given any node such that everything above it already has a label (for example, 0, 2, 0), we first arrange these in decreasing order (2, 0, 0), raise ω to the power of each these labels (ω^2, 1, 1; notice that $\omega^0=1$), and add them together in the given order ($\omega^2 + 2$). This process is repeated until all the nodes have labels; the label at the root is then taken as the ordinal for the hydra as a whole.

As an exercise, you might like to check that the hydra shown in Error: Reference source not found has the ordinal:

$$\omega^{\omega^{\omega^{2}+2}+1} + \omega^{\omega+1} + 1$$

Looking again at the rules of the battle as described in Chapter 6, it's now not too hard to convince oneself that each round decreases the ordinal associated with the hydra, since the loss of the head in question more than compensates for the effect of all the newly added portions.

[34] OK, here's how far. The ordinal ε_0 we've just reached is what's called the first ε-*number*. An ε-number is any ordinal α such that $\alpha = \omega^{\alpha}$. To reach the second ε-number (written as ε_1), we must take the limit of the sequence ε_0, ω^{ε_0}, $\omega^{\omega^{\varepsilon_0}}$, …. And so on.

Can we get as far as ε_1 using PA_1 as our base system? Sure we can. And we can get to ε_2, ε_3, …, ε_ω, …, and even as far as $\varepsilon_{\varepsilon_0}$, the ε_0-th ε-number. And that's only the beginning: we can get to $\varepsilon_{\varepsilon_{\varepsilon_0}}$, and indeed we can continue this nesting of 'ε' subscripts to any finite depth we like.

But now what about the limit of the sequence 0, ε, $\varepsilon_{\varepsilon_0}$, $\varepsilon_{\varepsilon_{\varepsilon_0}}$, …? It's at this point that we need to step up to PA_2 in order to show that the limit is indeed well-founded. So the girls can travel a quite colossal distance before our tortoise makes even its second step. (And as you would by now expect, the distance before the tortoise's third step is 'even more colossal'.)

What we've described here amounts, more or less, to just the first two levels of the so-called *Veblen hierarchy*: a super-stupendous construction used to map out the ordinals that come within reach as the tortoise climbs the existing staircase. Don't even *try* to imagine it!

[35] The general concept of an autonomous progression was introduced by Georg Kreisel in 1958 (cf. endnote [?]), and further developed by Solomon Feferman in the early 1960s.

[36] See S. Feferman, *Systems of predicative analysis*, Journal of Symbolic Logic 29(1), 1-30 (1964), and K. Schütte, *Predicative well-orderings*, in J. Crossley and M. Dummett, eds, *Formal systems and recursive functions*, North-Holland (1965). For textbook presentations of this material, see Schütte (1977) and Pohlers (2009).

[37] See for example S. Feferman, *Predicativity*, in Shapiro (2005), and Chapter 22 of Halbach (2010). For an alternate (though not widely accepted) point of view, see N. Weaver, *Predicativity beyond* Γ_0 (2005), available online.

[38] Two such systems, whose definition makes no reference to Γ_0 but whose first-order theorems coincide exactly with those of PA_{Γ_0}, are Feferman's system

IR of *inductive-recursive analysis*, and the system ATR_0 embodying *arithmetical transfinite recursion*. See Simpson (1999).

How plausible is it that even one of these simpler systems, or something like it, is in some way wired into our brains? I really don't know, so I'll leave this as a topic for discussion.